

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

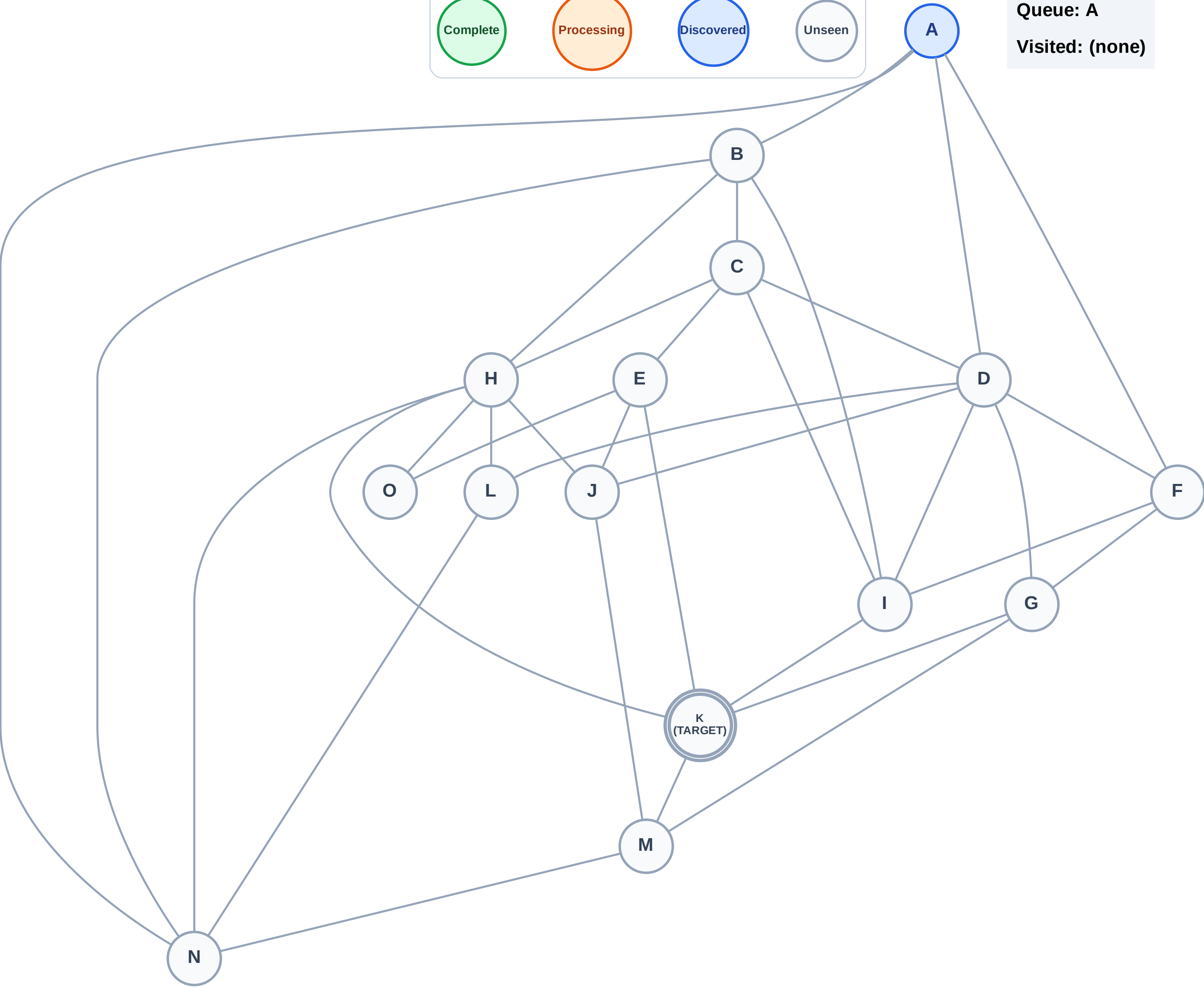
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



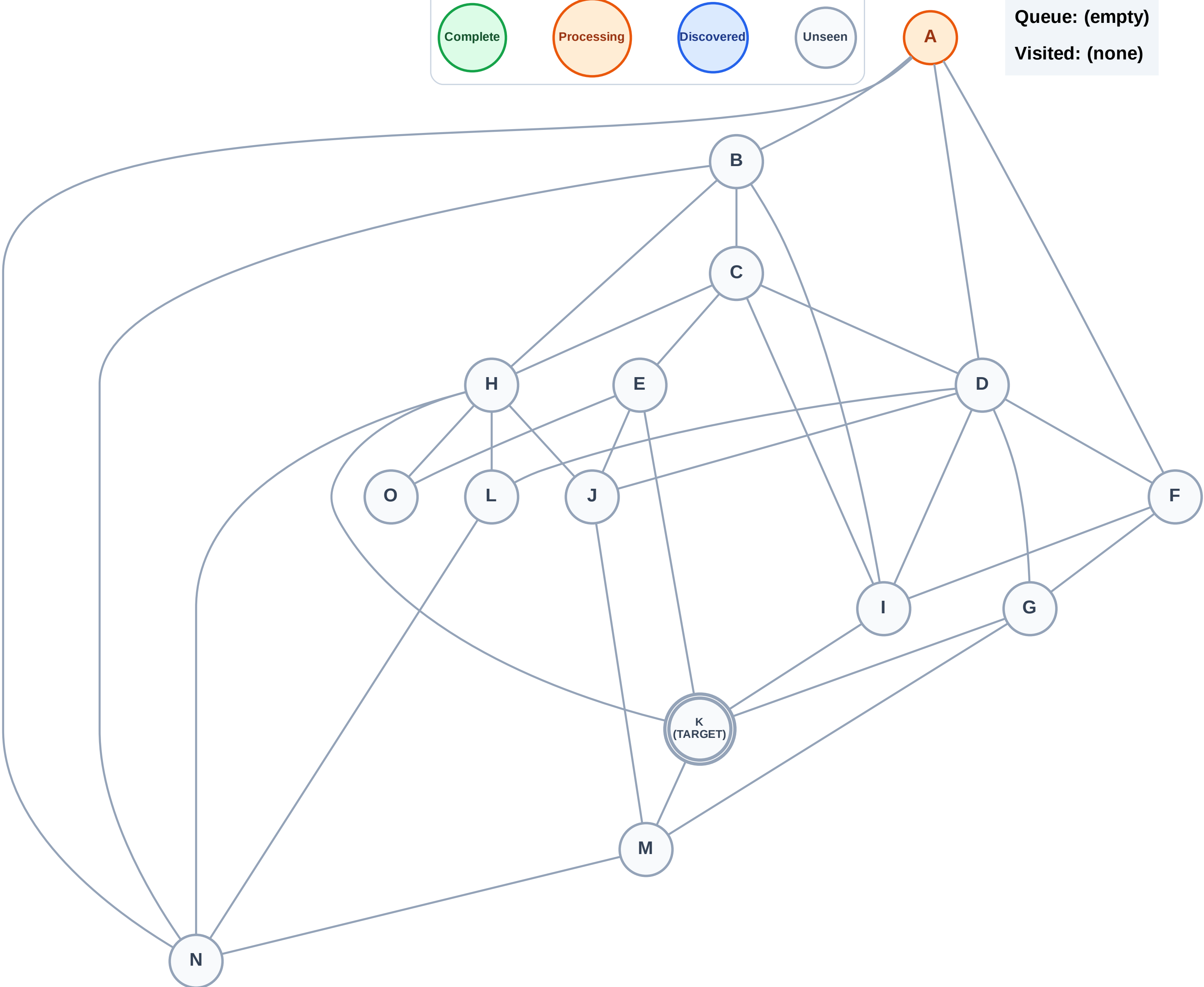
BFS Traversal

Step 2: Process node A

Legend



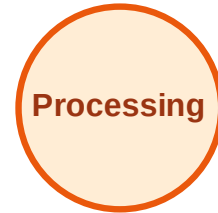
Queue: (empty)
Visited: (none)



BFS Traversal

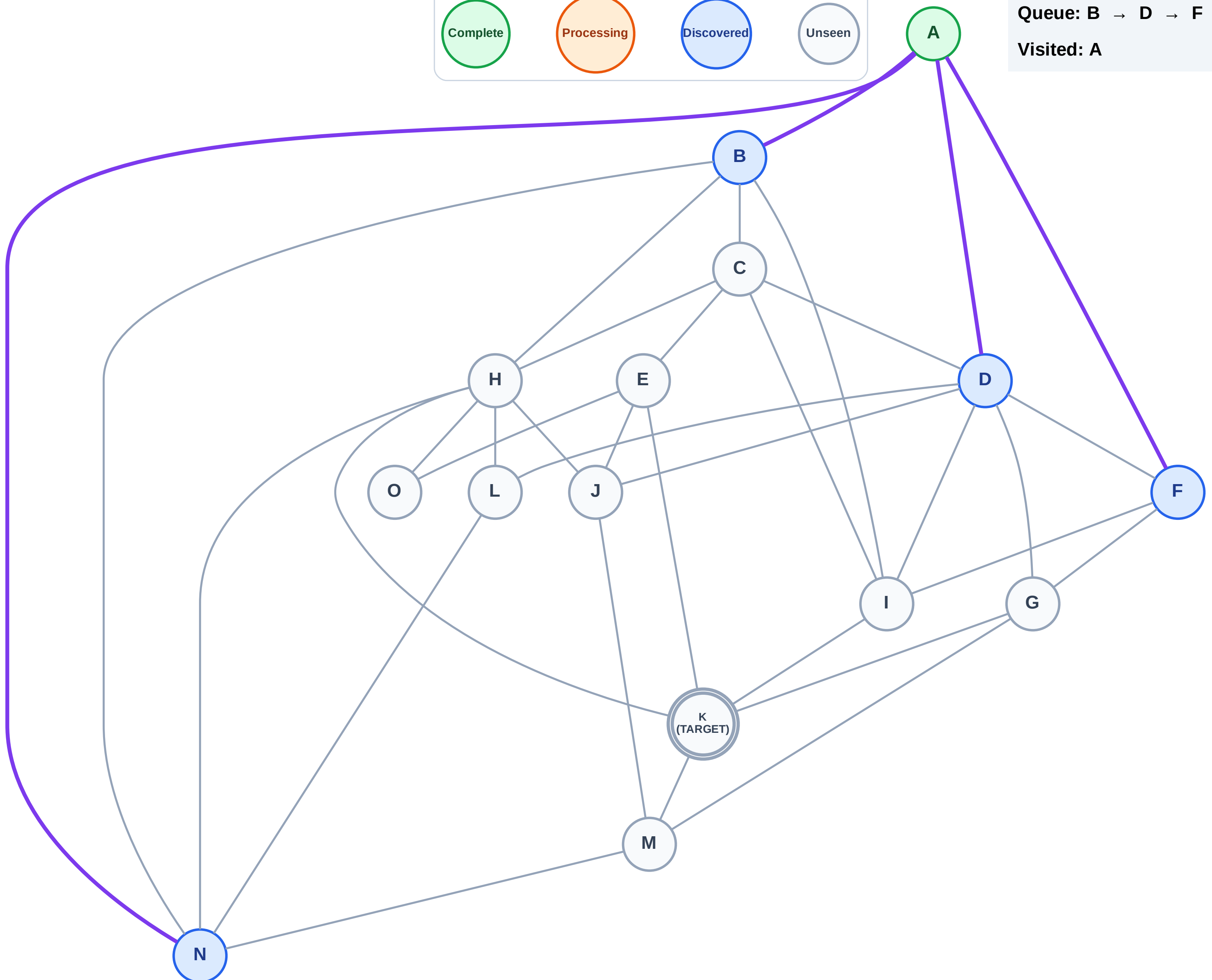
Step 3: Discover B, D, F, N from A

Legend



Queue: B → D → F → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

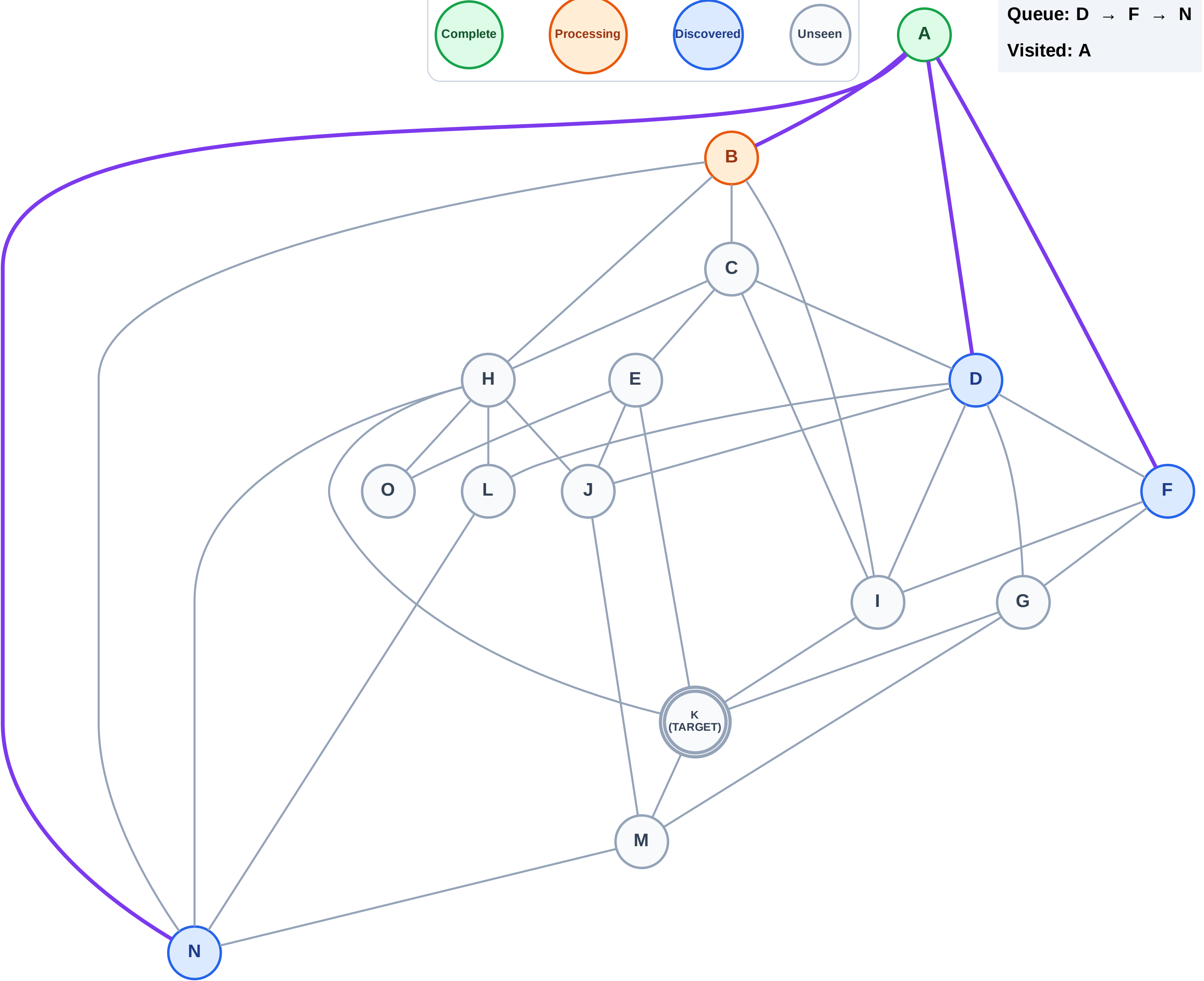
Processing

Discovered

Unseen

Queue: D → F → N

Visited: A



BFS Traversal

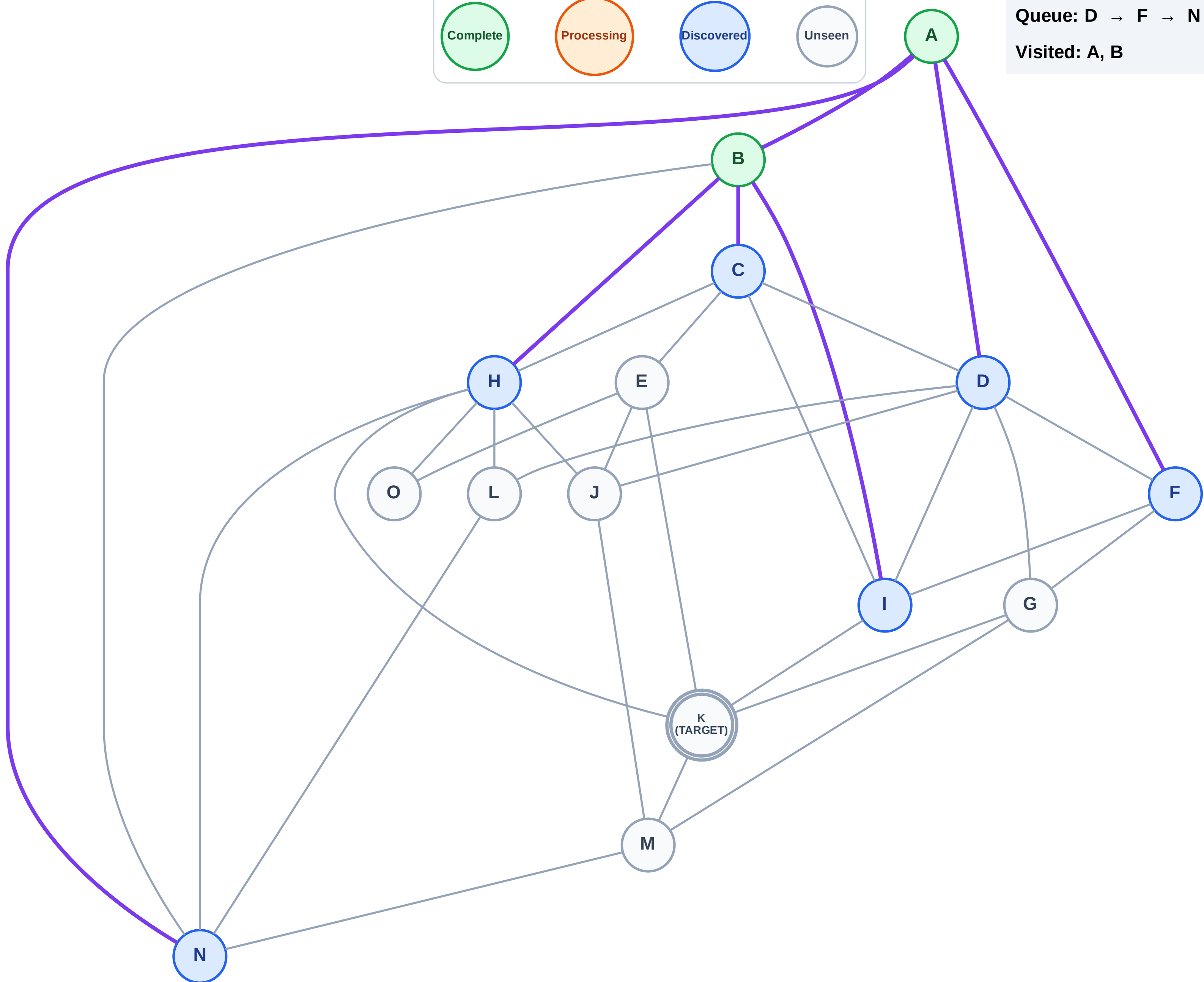
Step 5: Discover C, H, I from B

Legend



Queue: D → F → N → C → H → I

Visited: A, B



BFS Traversal

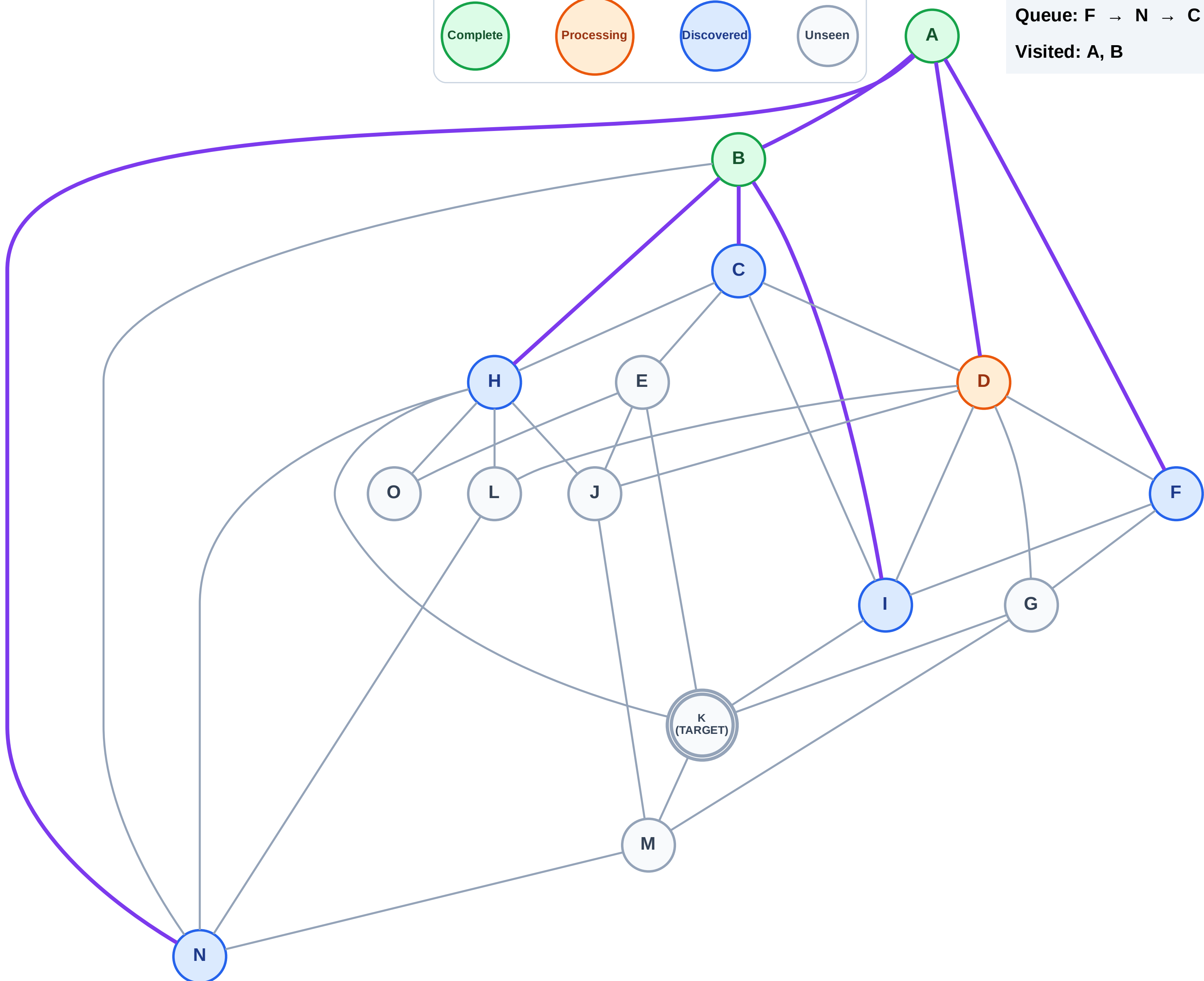
Step 6: Process node D

Legend



Queue: F → N → C → H → I

Visited: A, B



BFS Traversal

Step 7: Discover G, J, L from D

Legend



Complete



Processing



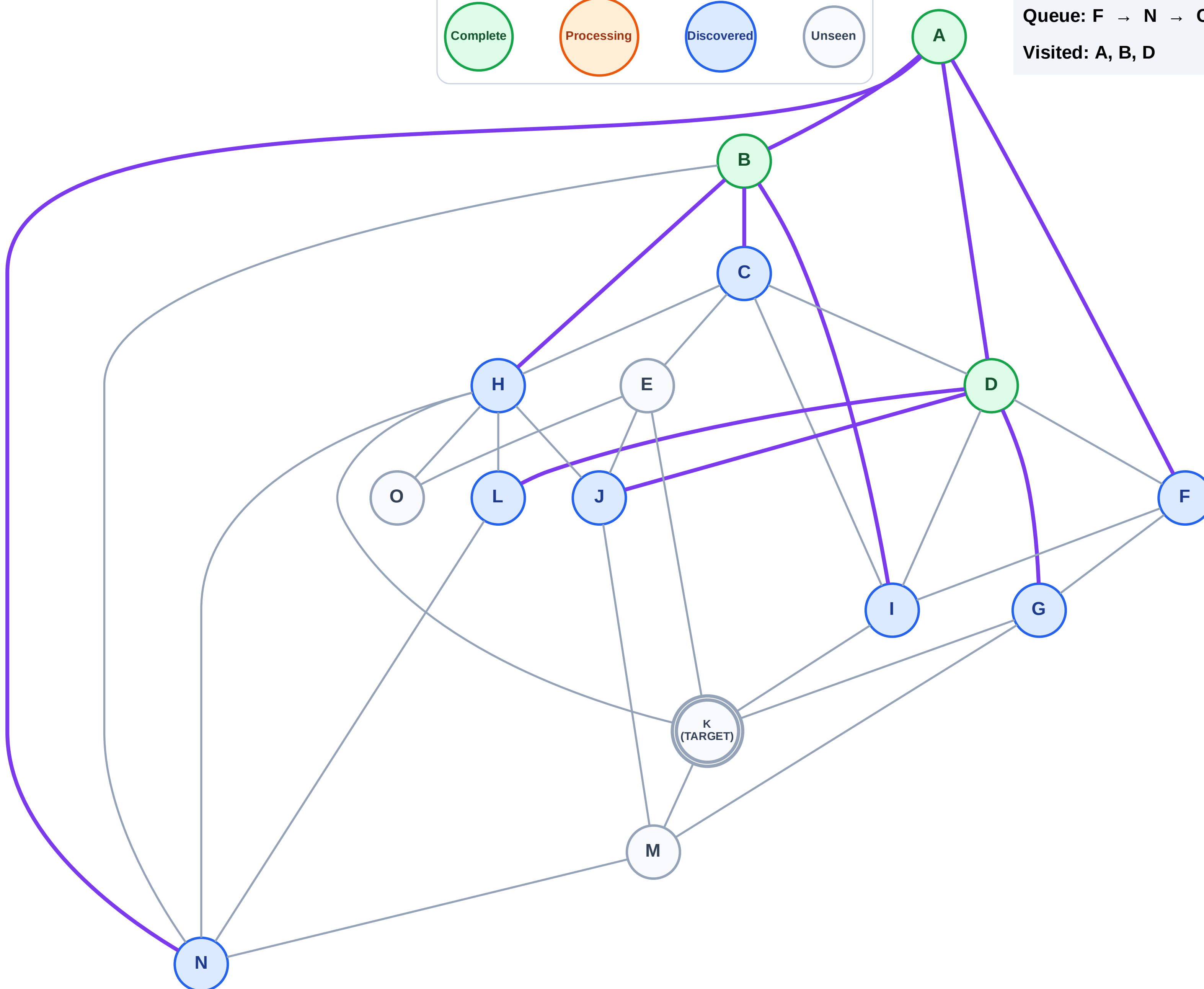
Discovered



Unseen

Queue: F → N → C → H → I → G → J → L

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

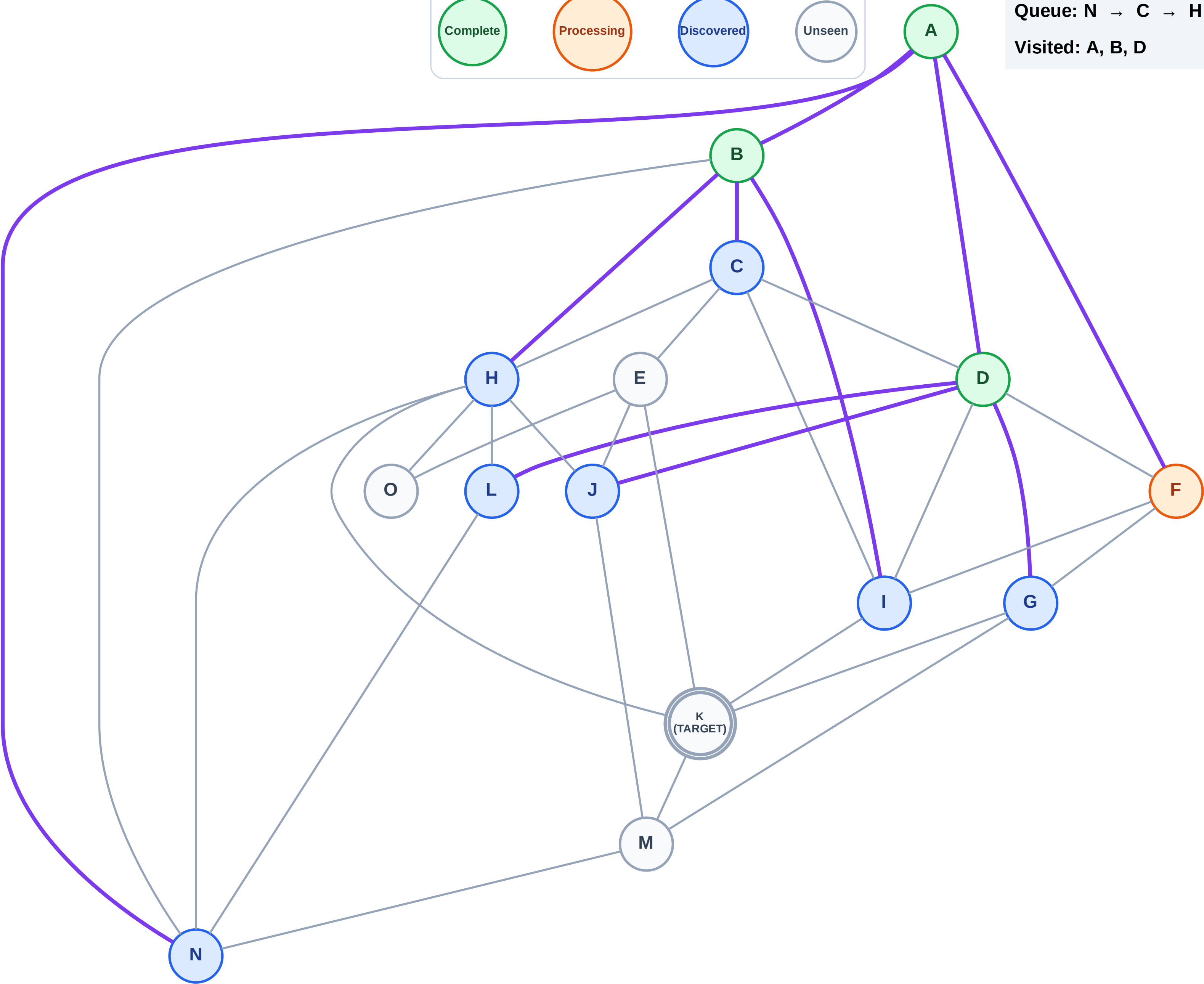
Processing

Discovered

Unseen

Queue: N → C → H → I → G → J → L

Visited: A, B, D



Step 9: No new neighbors from F

Step 9: No new neighbors from F

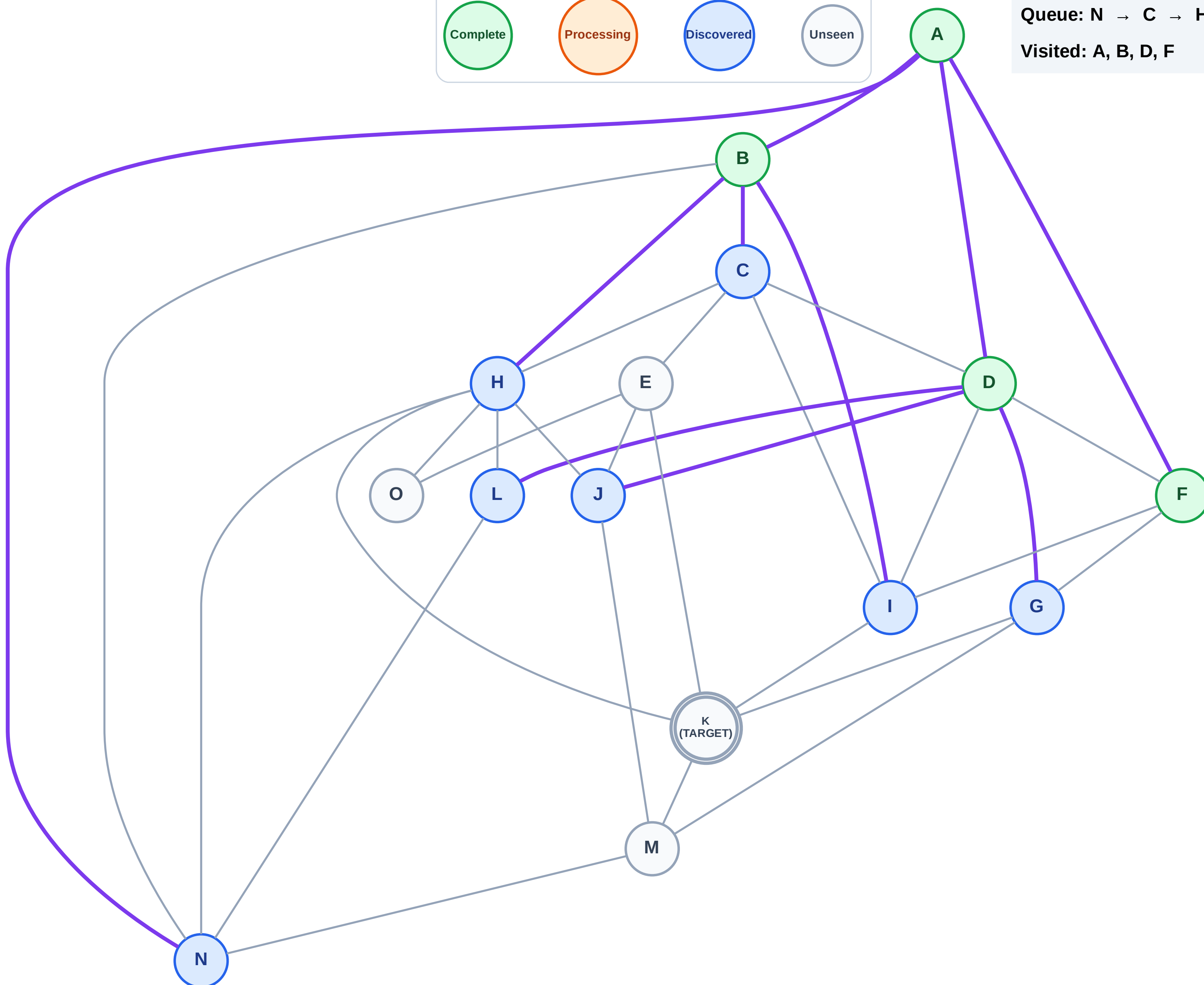
Legend

Complete Processing Discovered Unseen

Processing

Unseen

Visited: A, B, D, F



BFS Traversal

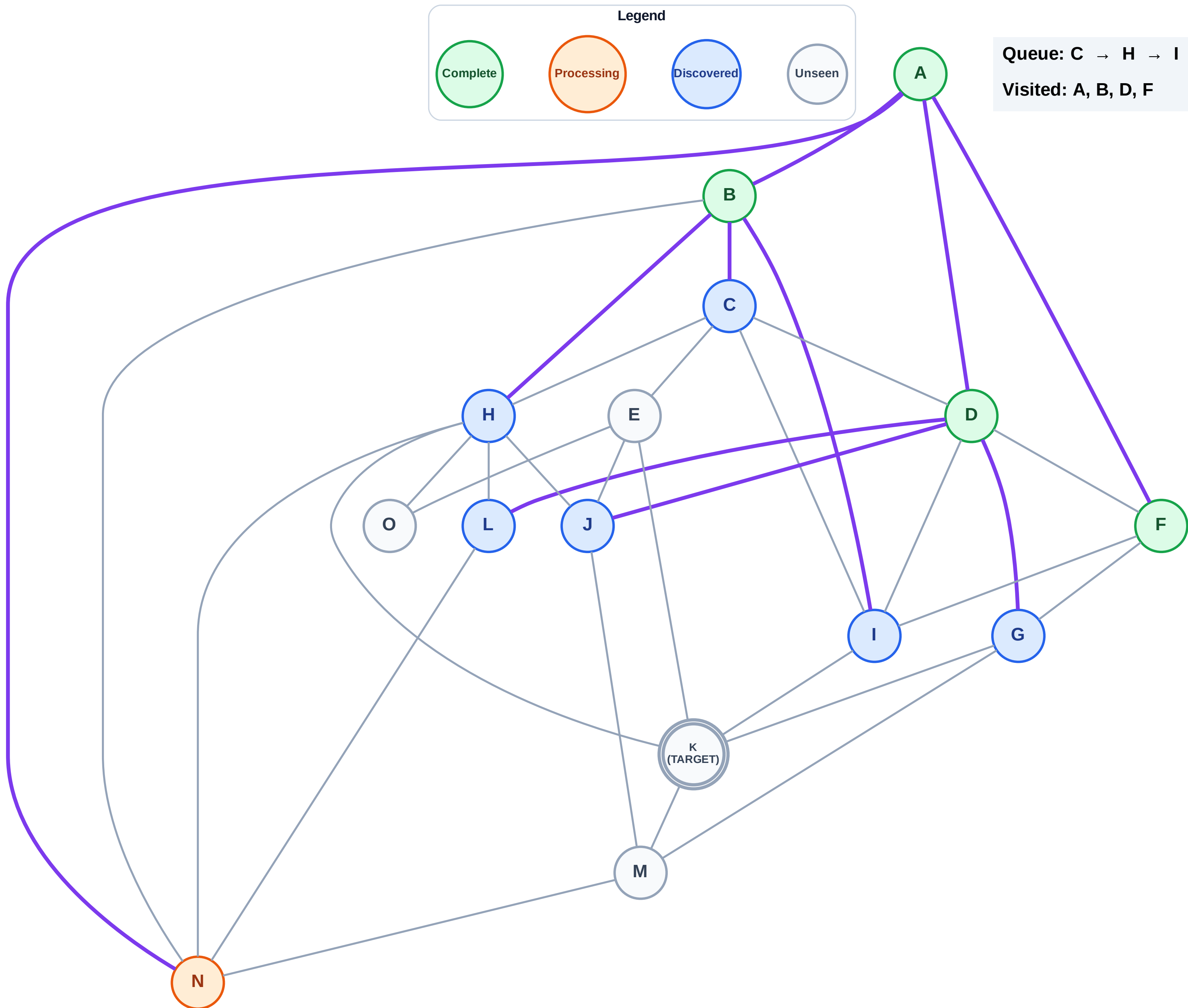
Step 10: Process node N

Legend



Queue: C → H → I → G → J → L

Visited: A, B, D, F



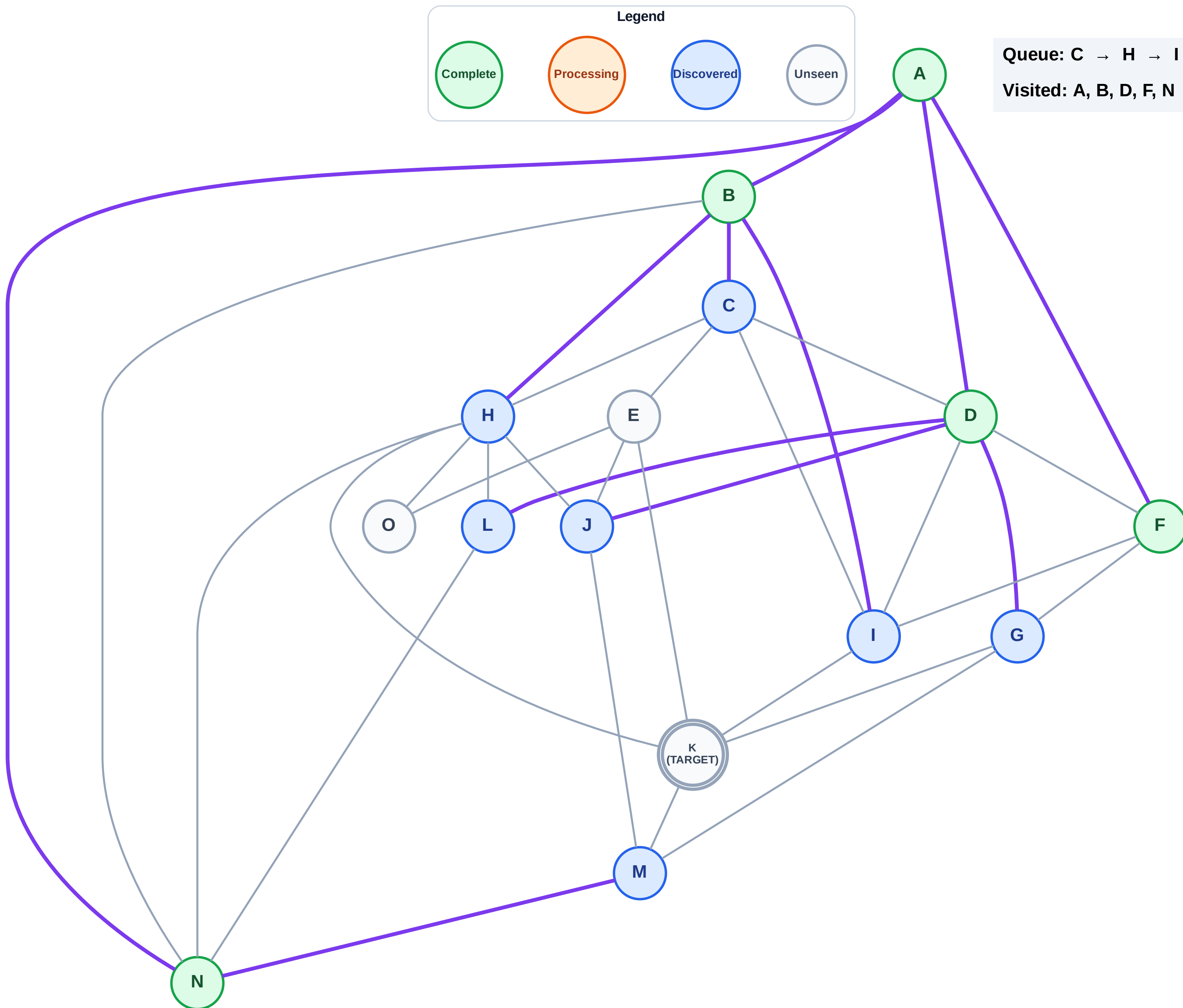
BFS Traversal

Step 11: Discover M from N

Legend



Queue: C → H → I → G → J → L → M
Visited: A, B, D, F, N



BFS Traversal

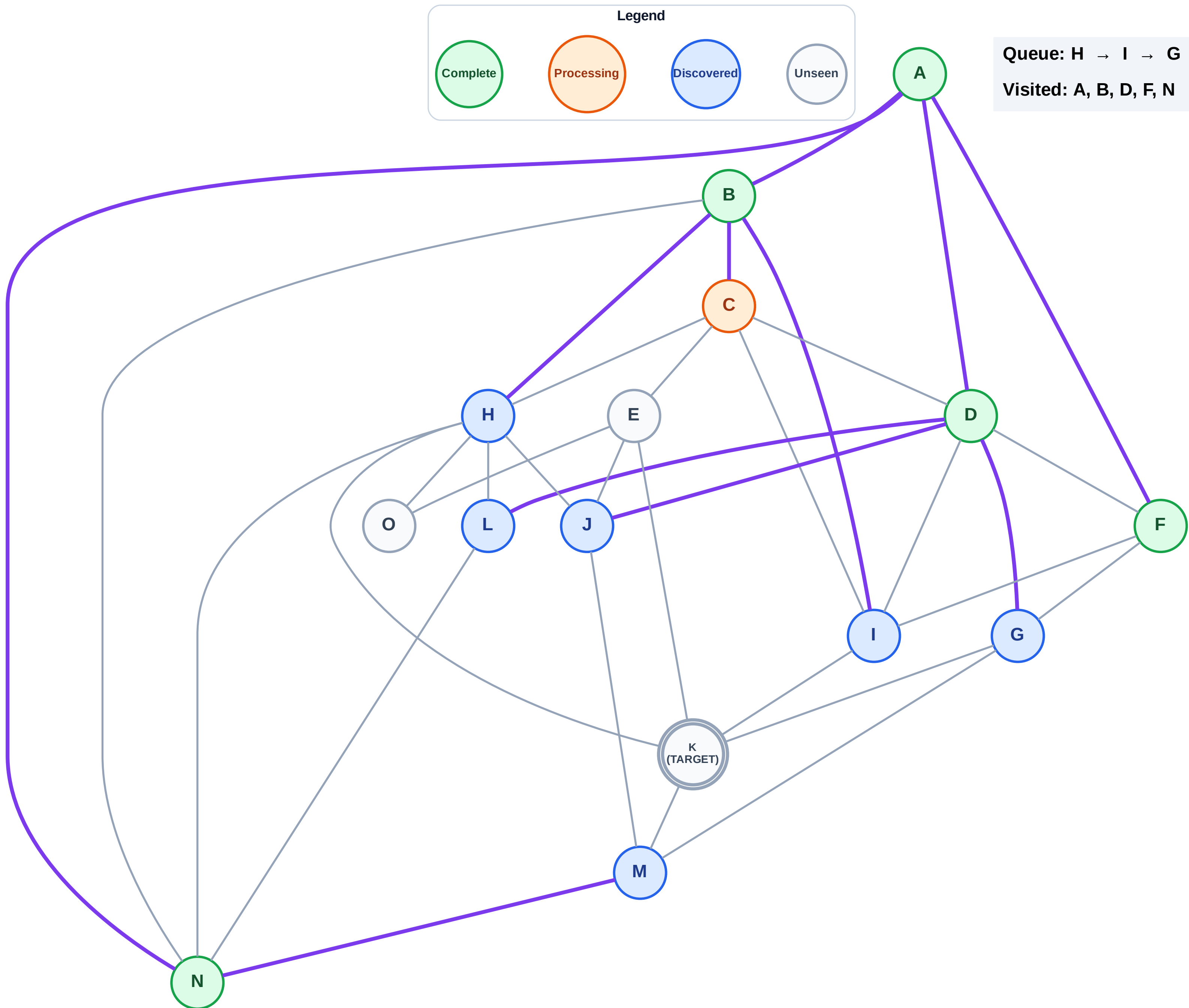
Step 12: Process node C

Legend



Queue: H → I → G → J → L → M

Visited: A, B, D, F, N



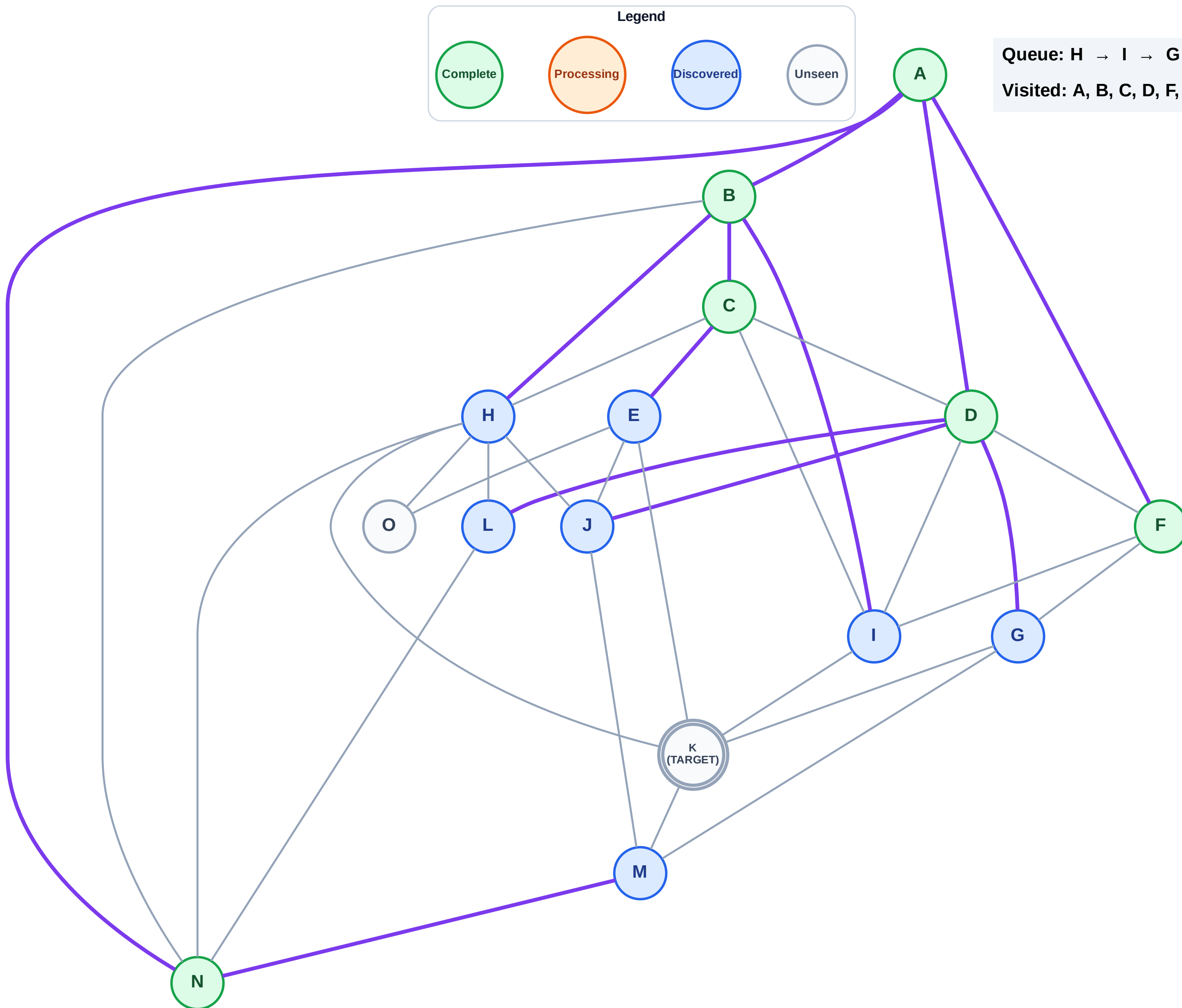
BFS Traversal

Step 13: Discover E from C

Legend



Queue: H → I → G → J → L → M → E
Visited: A, B, C, D, F, N



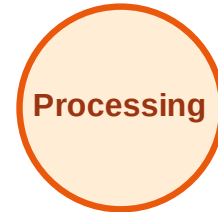
BFS Traversal

Step 14: Process node H

Legend



Complete



Processing



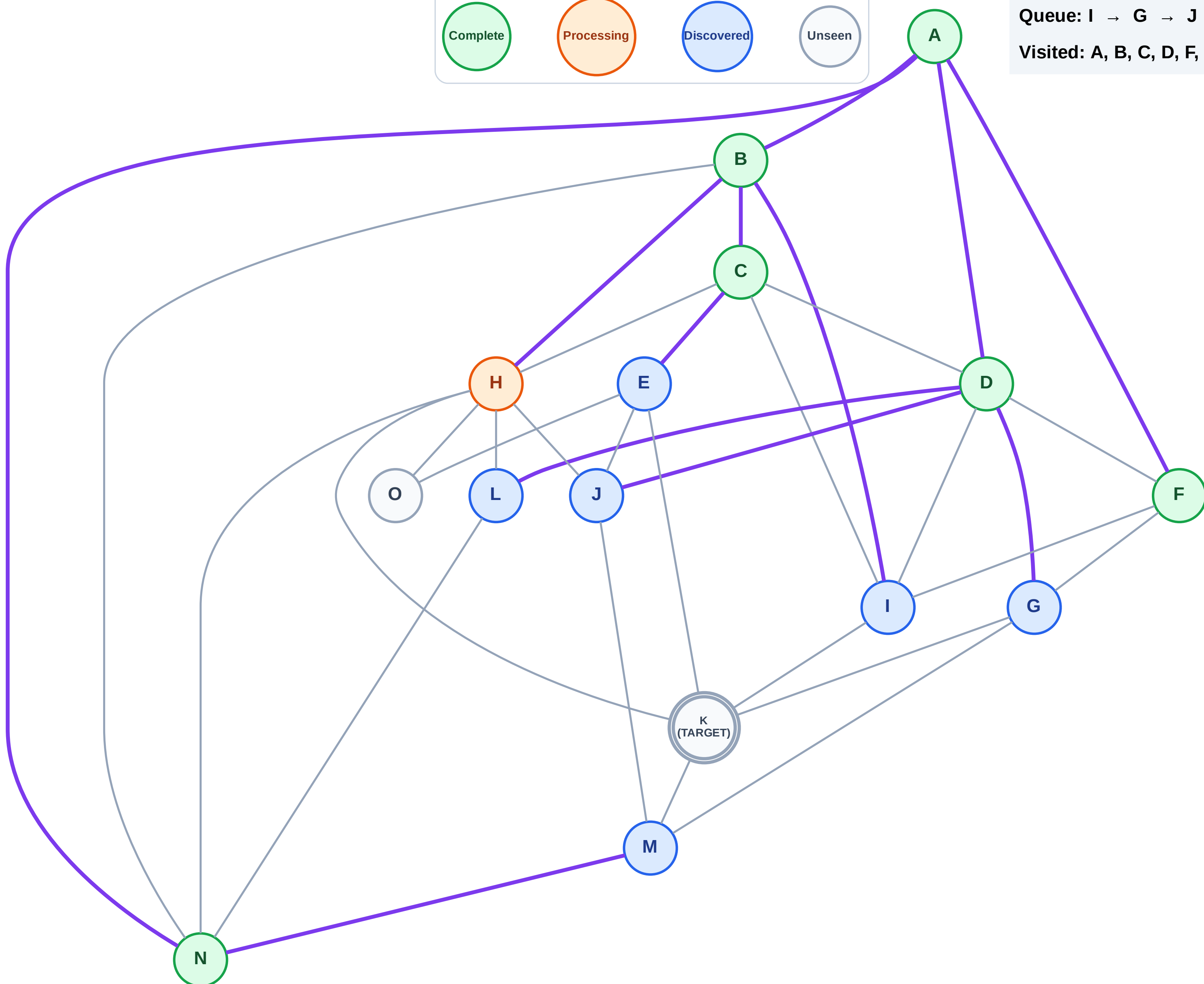
Discovered



Unseen

Queue: I → G → J → L → M → E

Visited: A, B, C, D, F, N



BFS Traversal

Step 15: Discover K, O from H

Legend



Complete



Processing



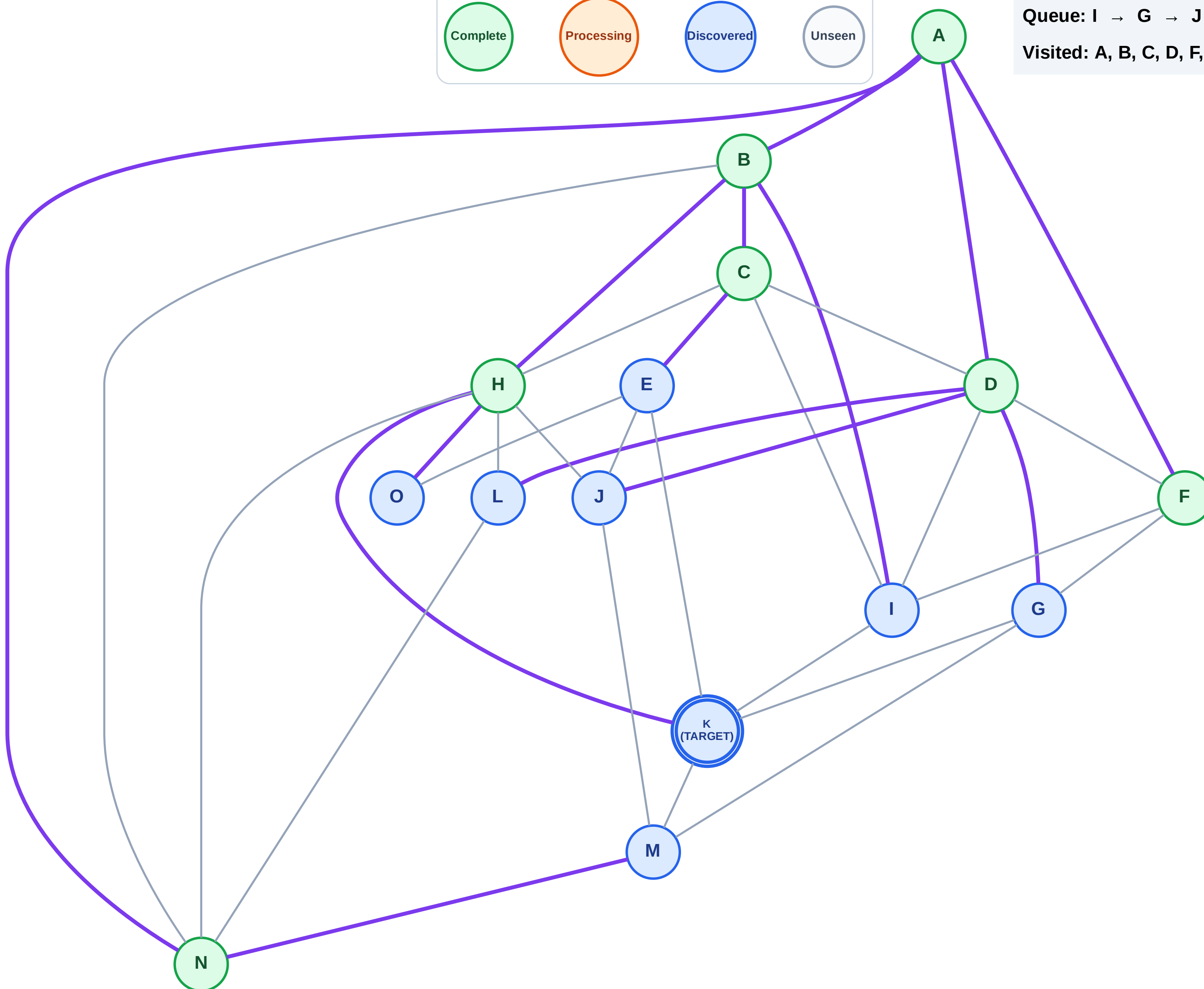
Discovered



Unseen

Queue: I → G → J → L → M → E → K → O

Visited: A, B, C, D, F, H, N



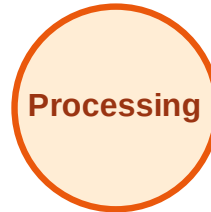
BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



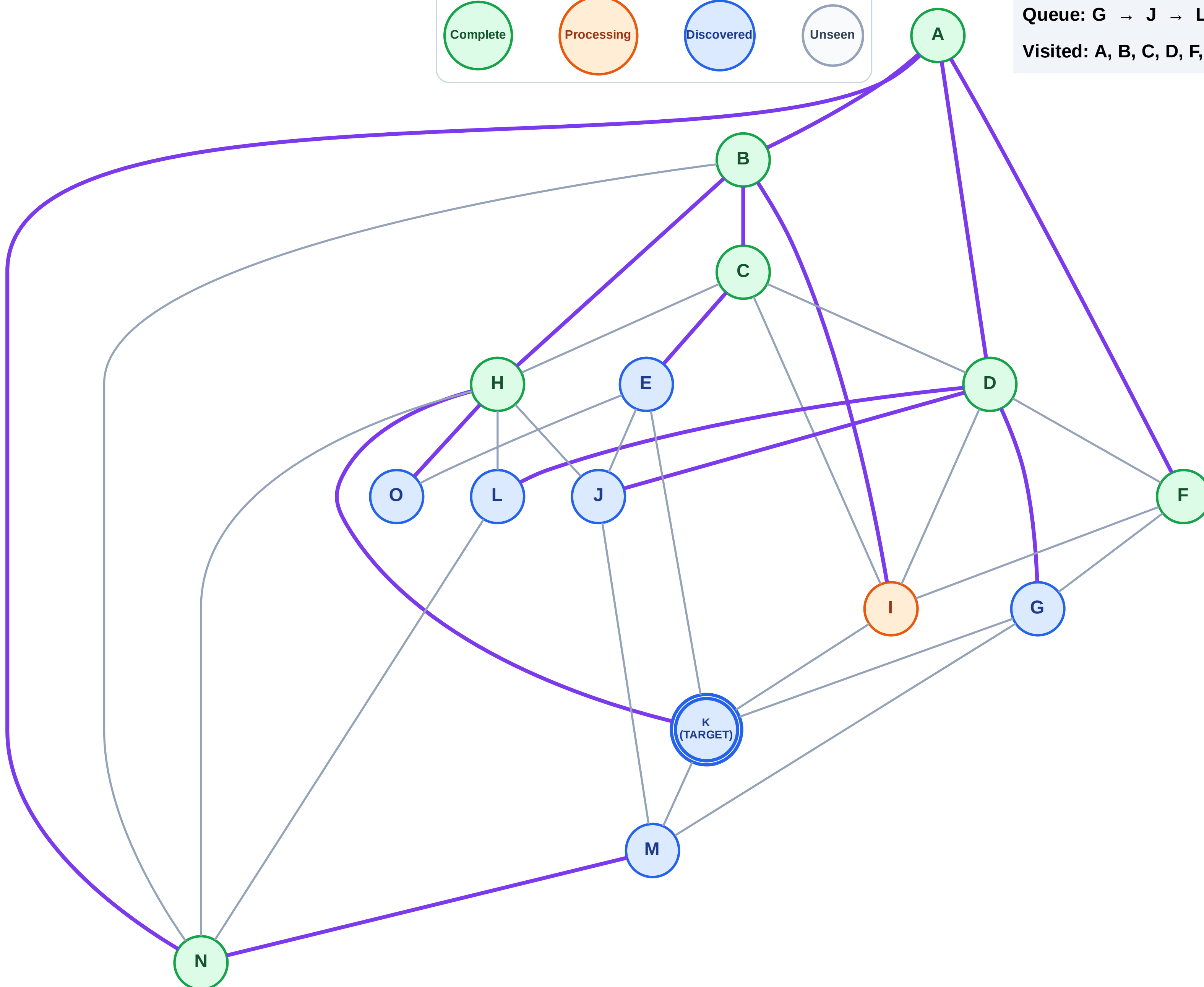
Discovered



Unseen

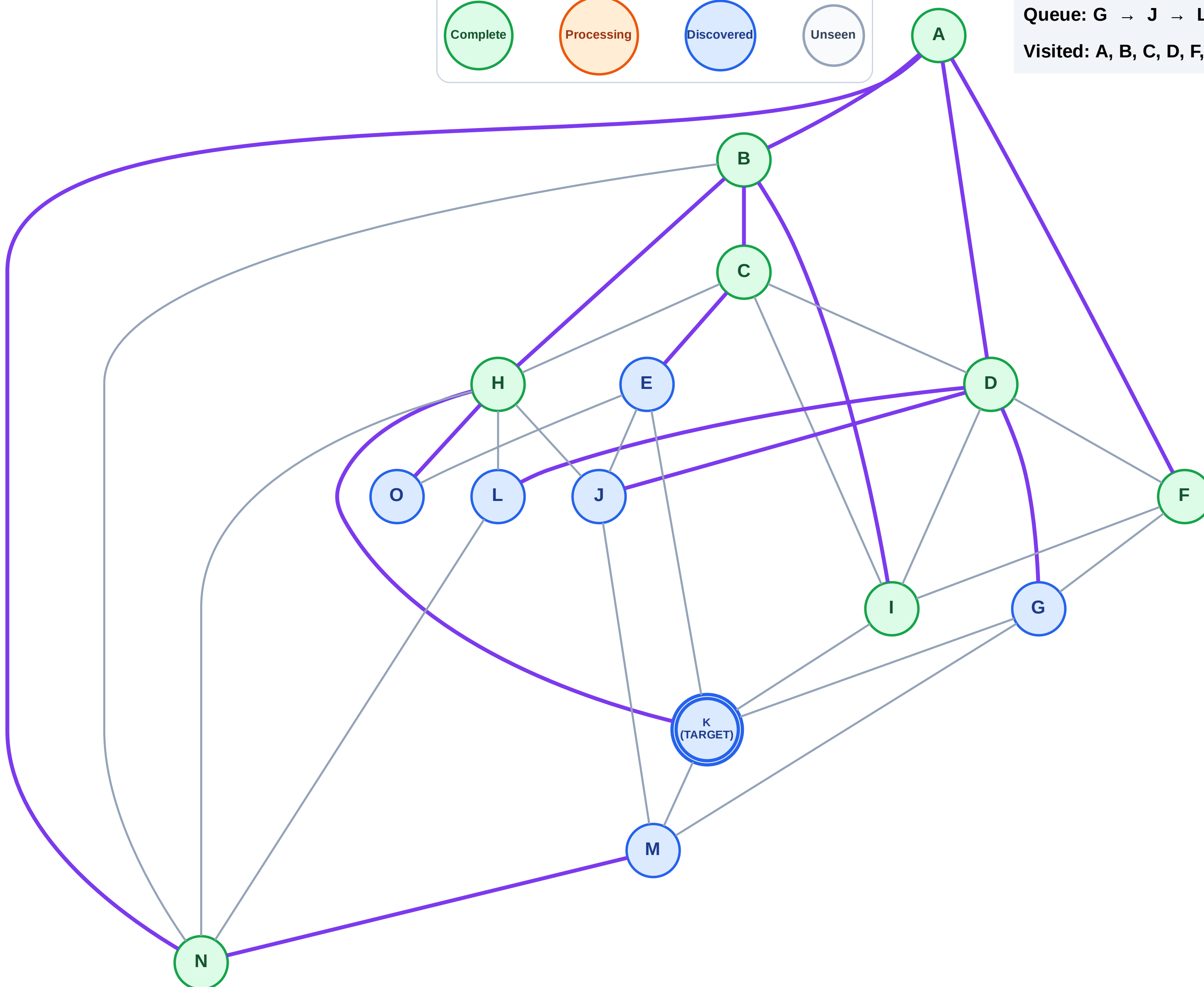
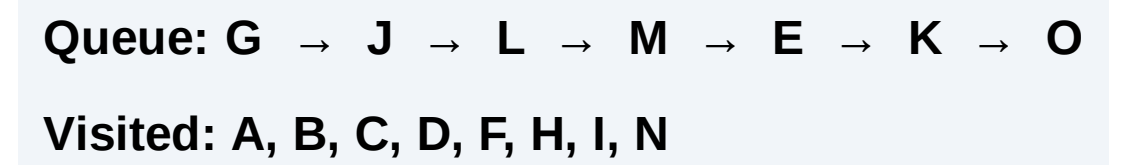
Queue: G → J → L → M → E → K → O

Visited: A, B, C, D, F, H, N



Step 17: No new neighbors from I

Legend

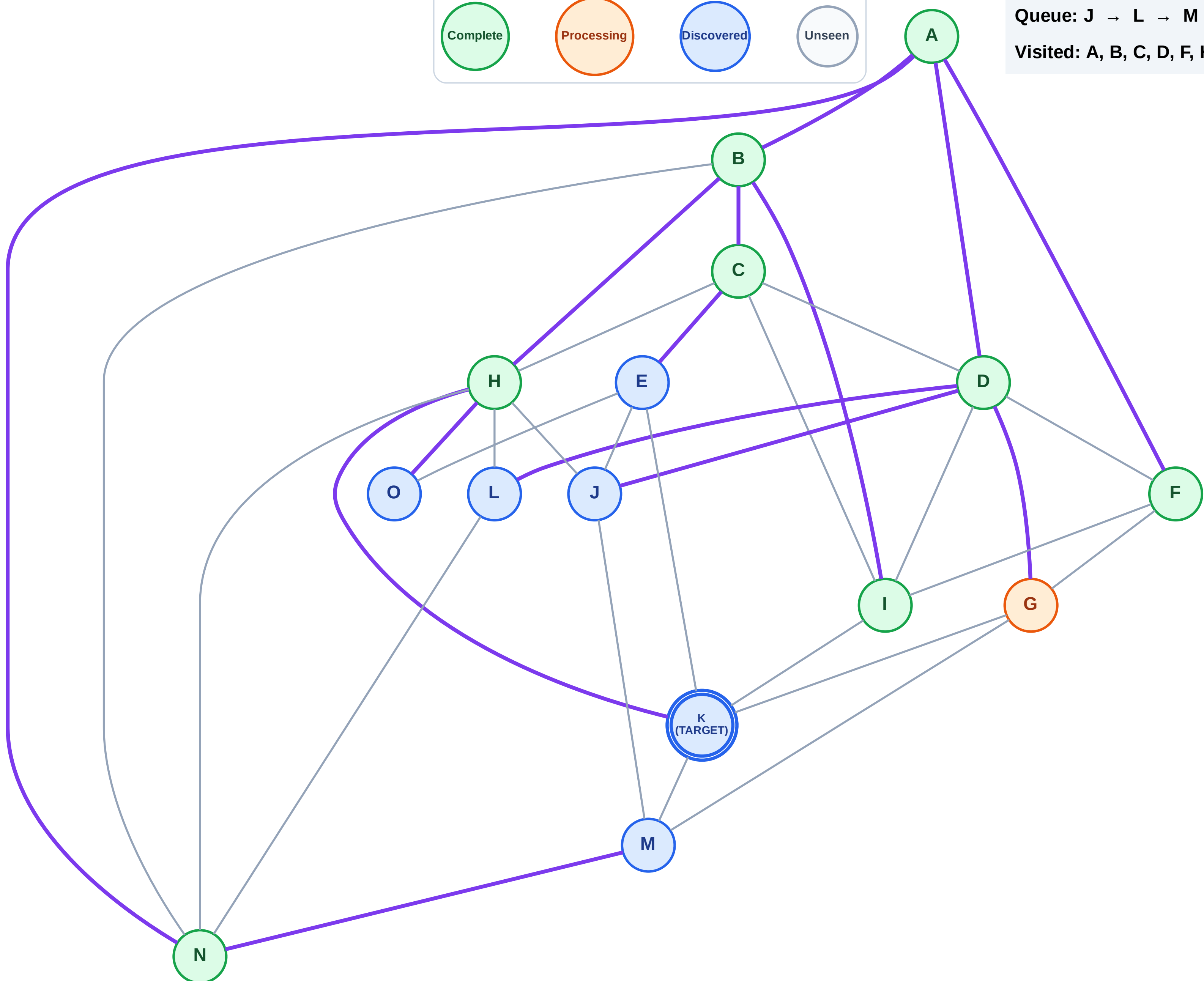


Step 18: Process node G

Complete

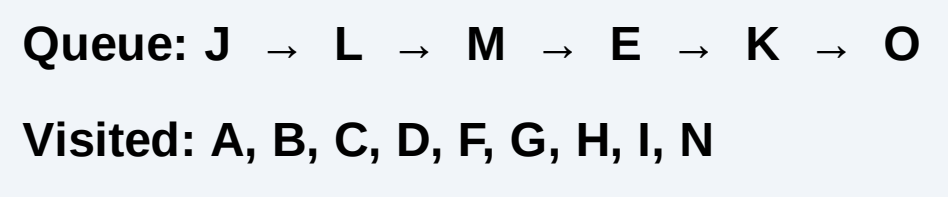


Visited: A, B, C, D, F, H, I, N



Step 19: No new neighbors from G

Step 19: No new neighbors from G



BFS Traversal

Step 20: Process node J

Legend

Complete

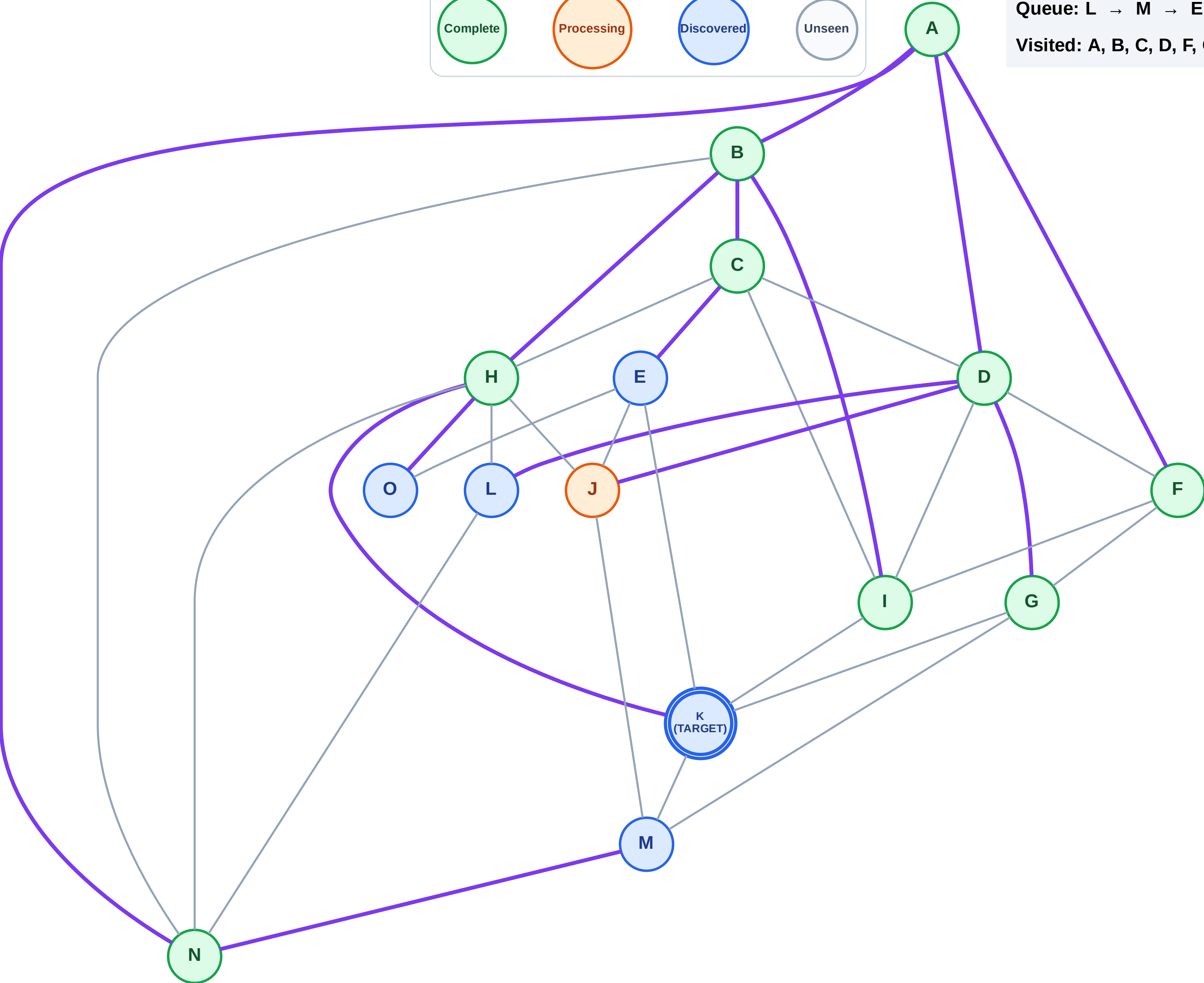
Processing

Discovered

Unseen

Queue: L → M → E → K → O

Visited: A, B, C, D, F, G, H, I, N



BFS Traversal

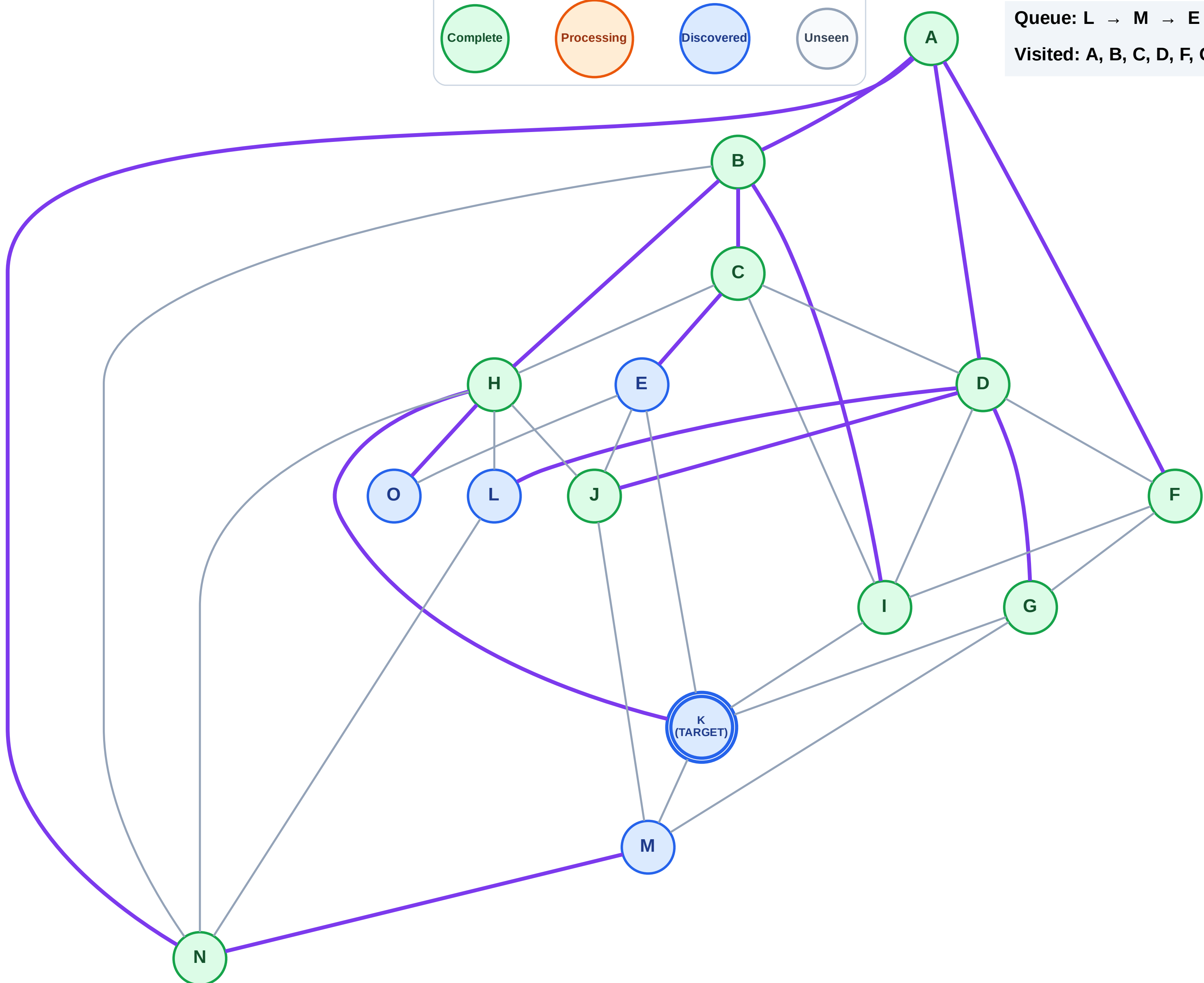
Step 21: No new neighbors from J

Legend



Queue: L → M → E → K → O

Visited: A, B, C, D, F, G, H, I, J, N



BFS Traversal

Step 22: Process node L

Legend

Complete

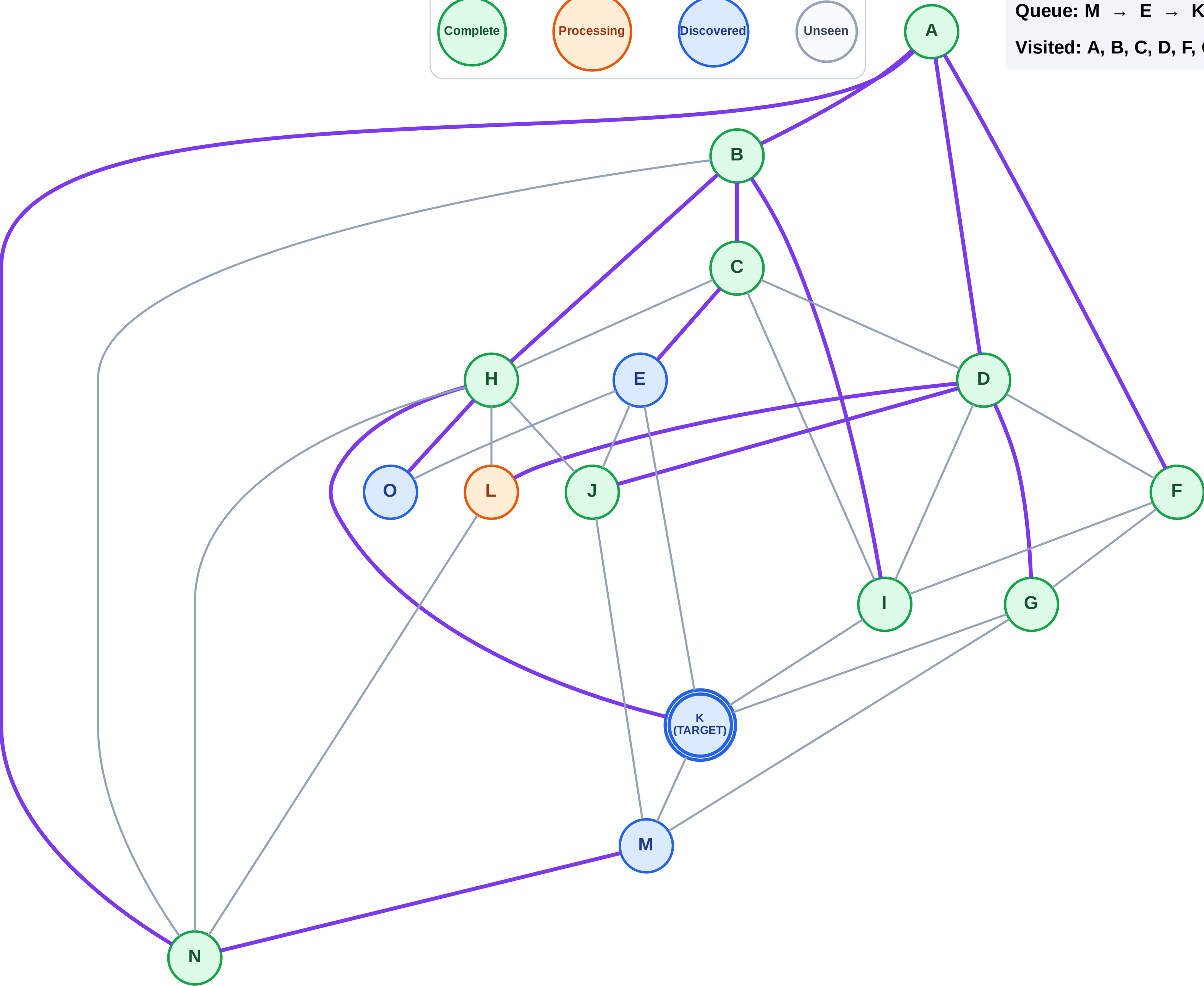
Processing

Discovered

Unseen

Queue: M → E → K → O

Visited: A, B, C, D, F, G, H, I, J, N



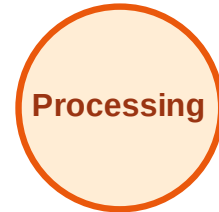
BFS Traversal

Step 23: No new neighbors from L

Legend



Complete



Processing



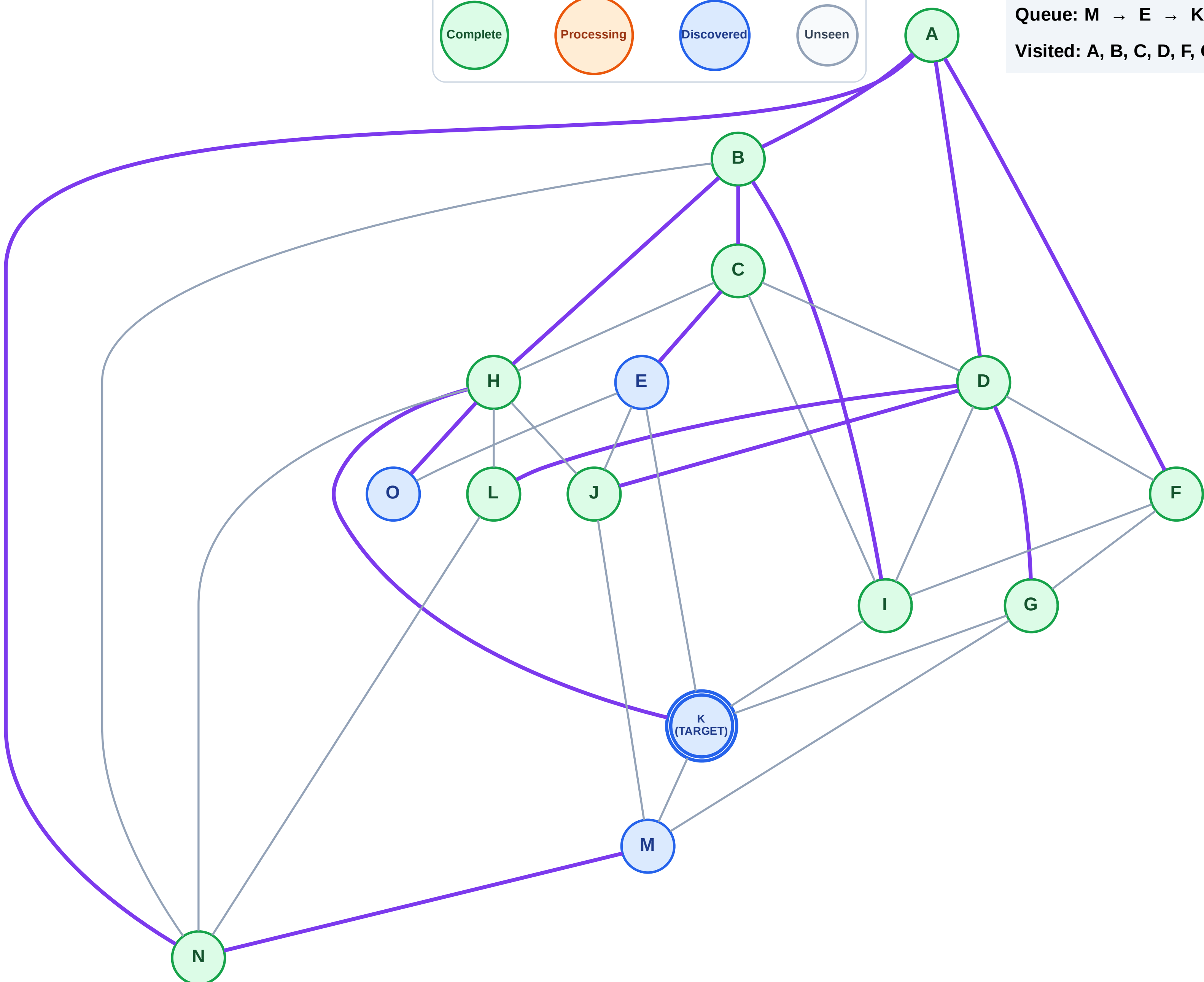
Discovered



Unseen

Queue: M → E → K → O

Visited: A, B, C, D, F, G, H, I, J, L, N



BFS Traversal

Step 24: Process node M

Legend

Complete

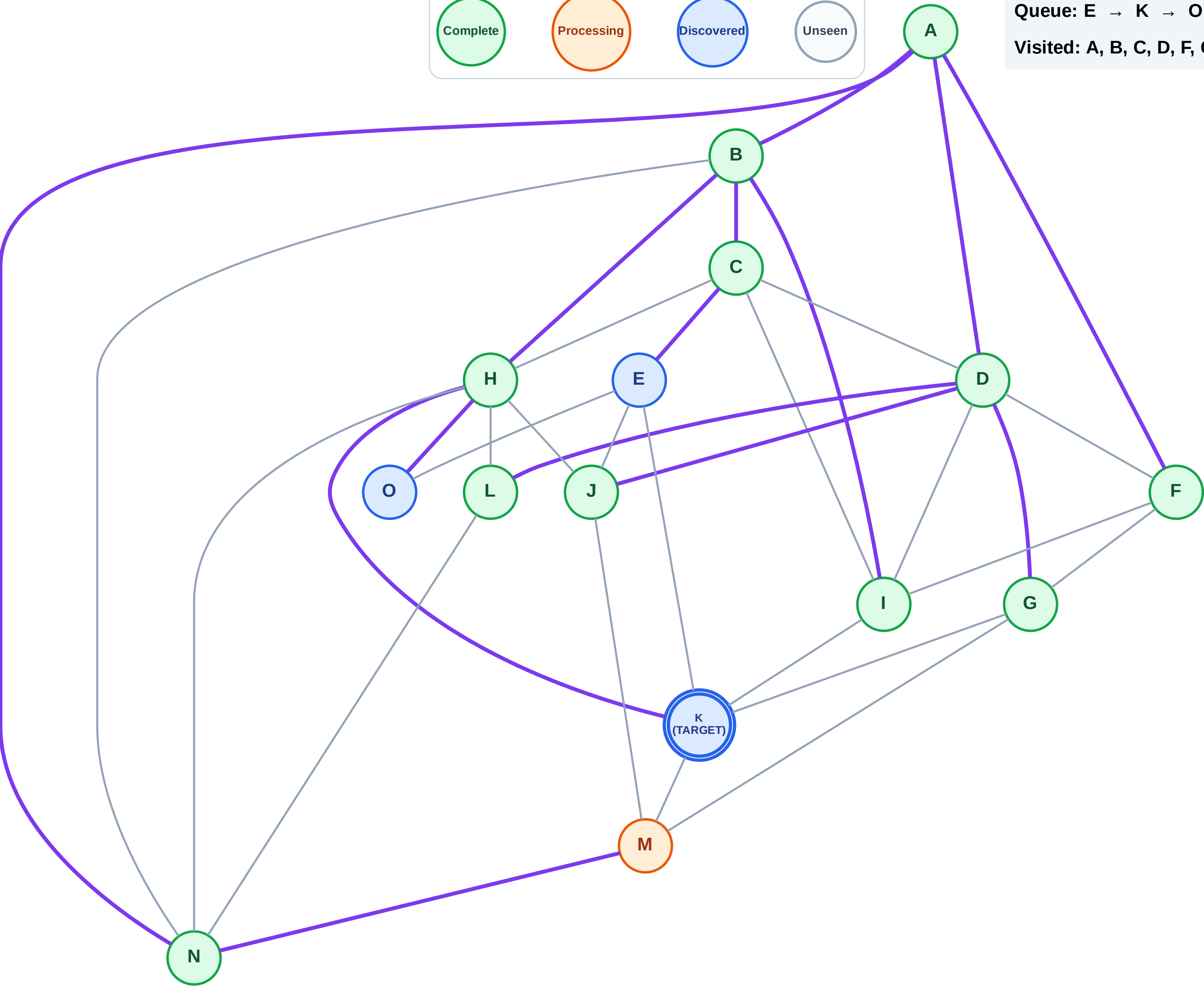
Processing

Discovered

Unseen

Queue: E → K → O

Visited: A, B, C, D, F, G, H, I, J, L, N



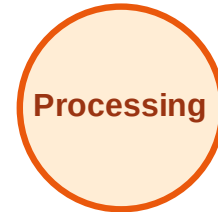
BFS Traversal

Step 25: No new neighbors from M

Legend



Complete



Processing



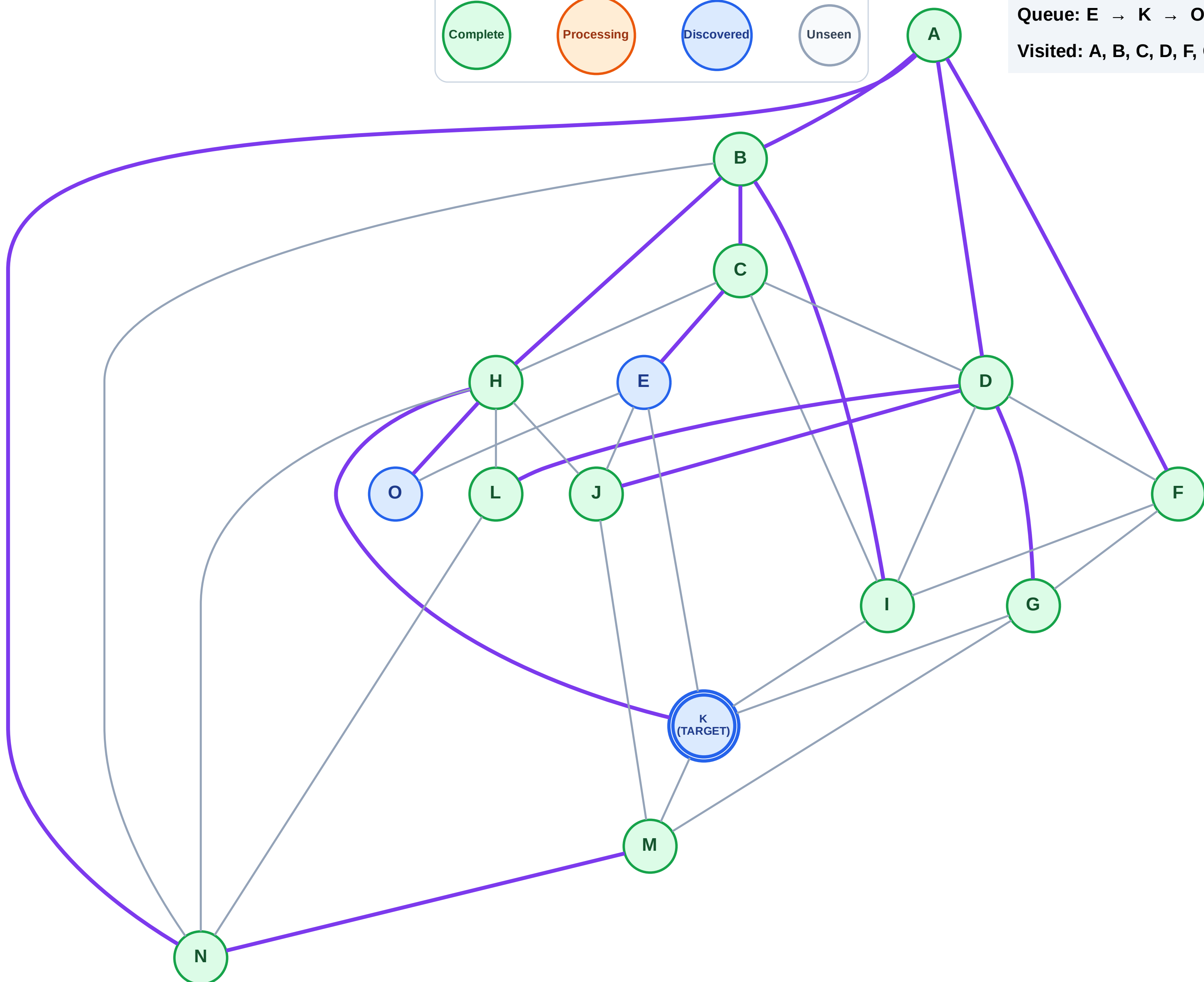
Discovered



Unseen

Queue: E → K → O

Visited: A, B, C, D, F, G, H, I, J, L, M, N



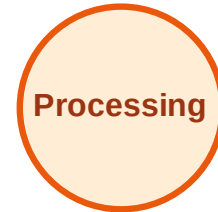
BFS Traversal

Step 26: Process node E

Legend



Complete



Processing



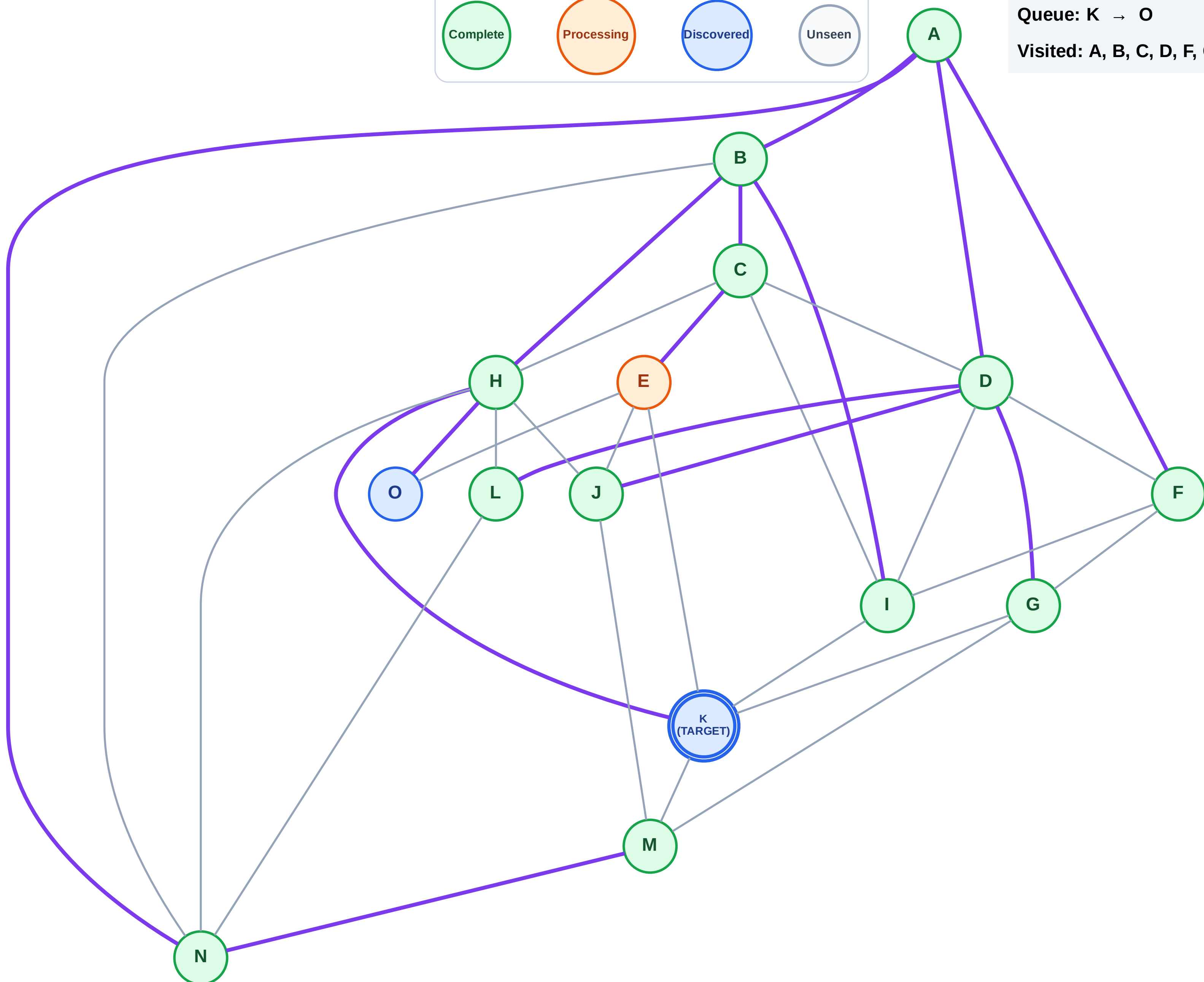
Discovered



Unseen

Queue: K → O

Visited: A, B, C, D, F, G, H, I, J, L, M, N



BFS Traversal

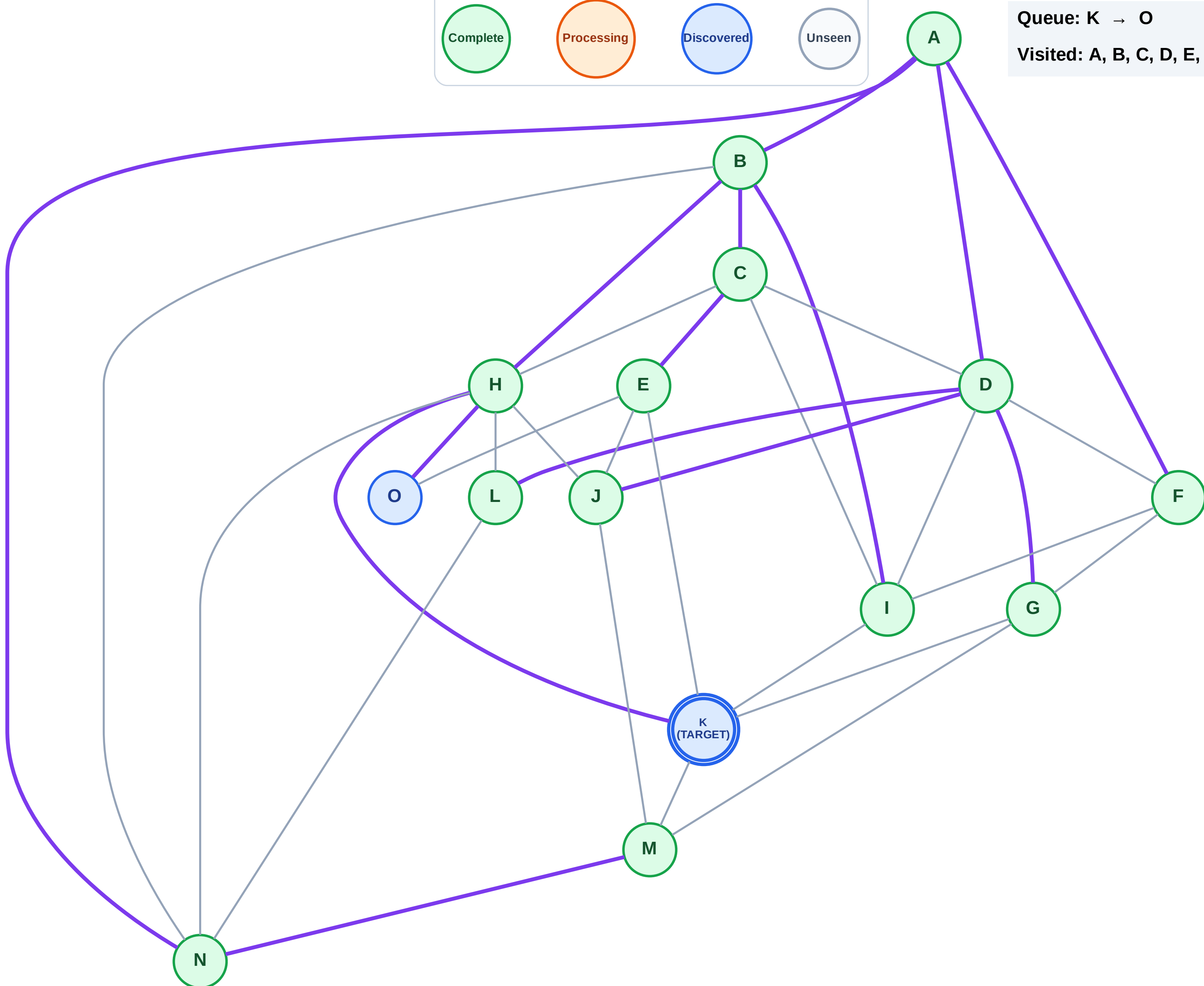
Step 27: No new neighbors from E

Legend



Queue: K → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



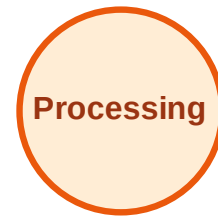
BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



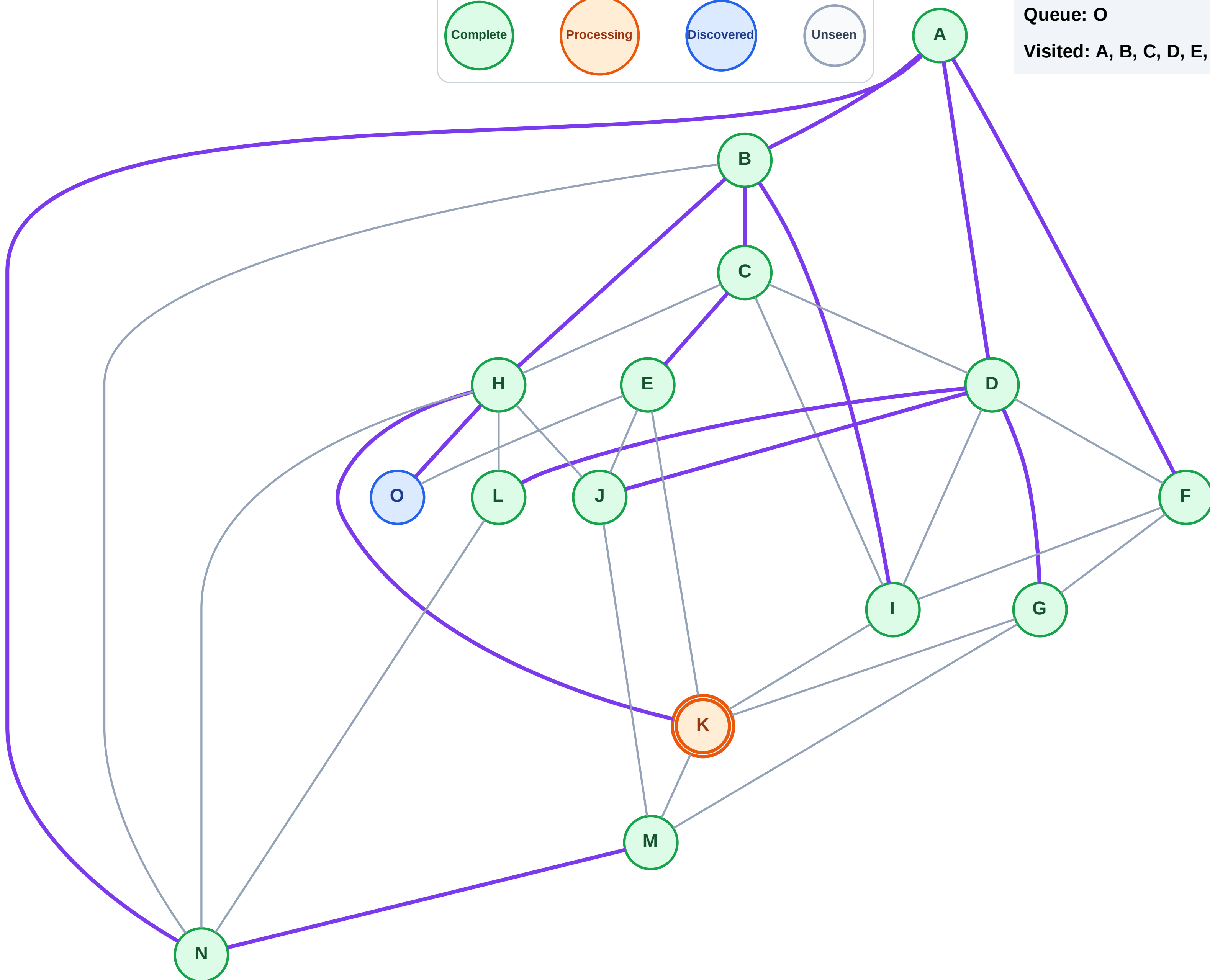
Discovered



Unseen

Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

Legend

Complete

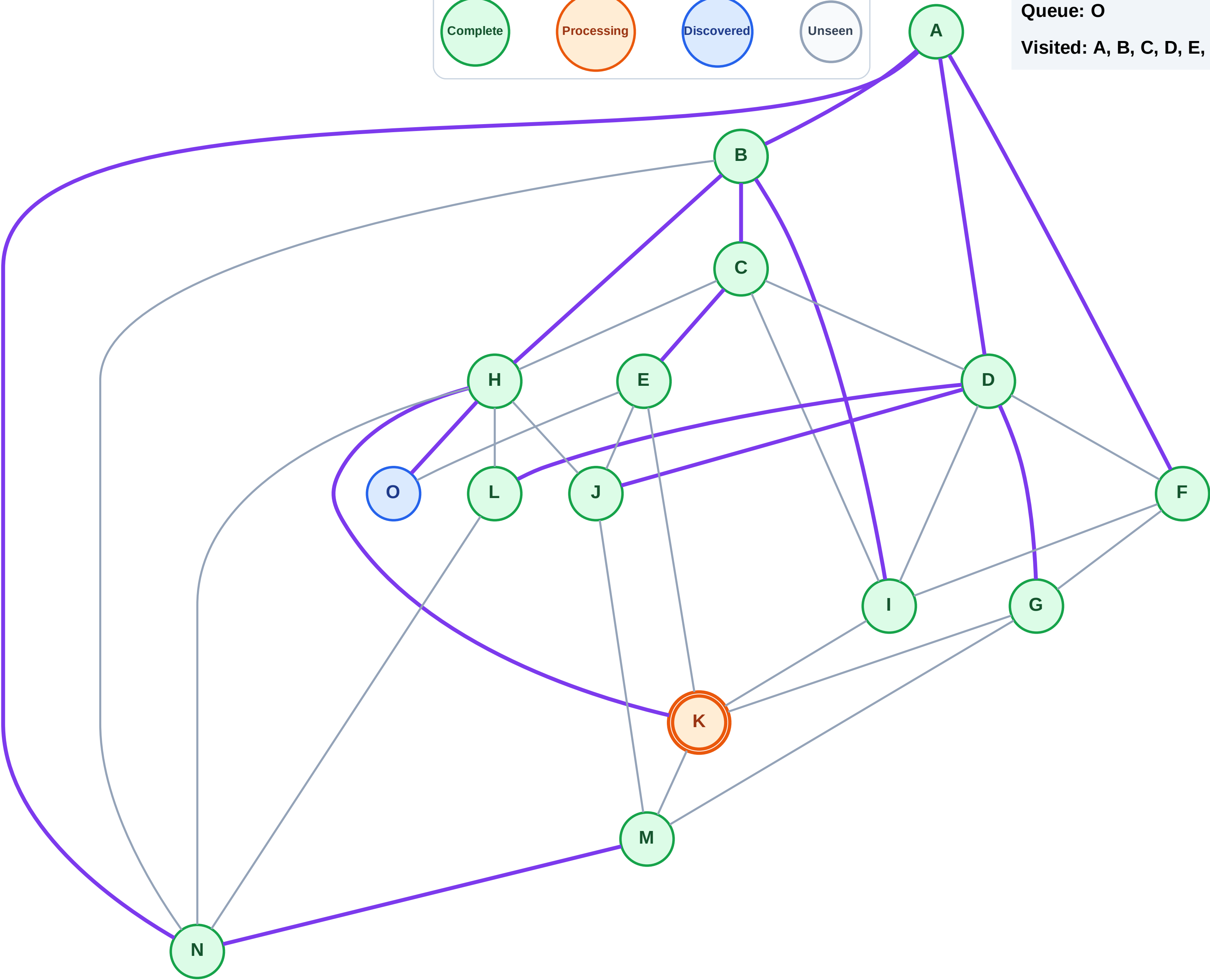
Processing

Discovered

Unseen

Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

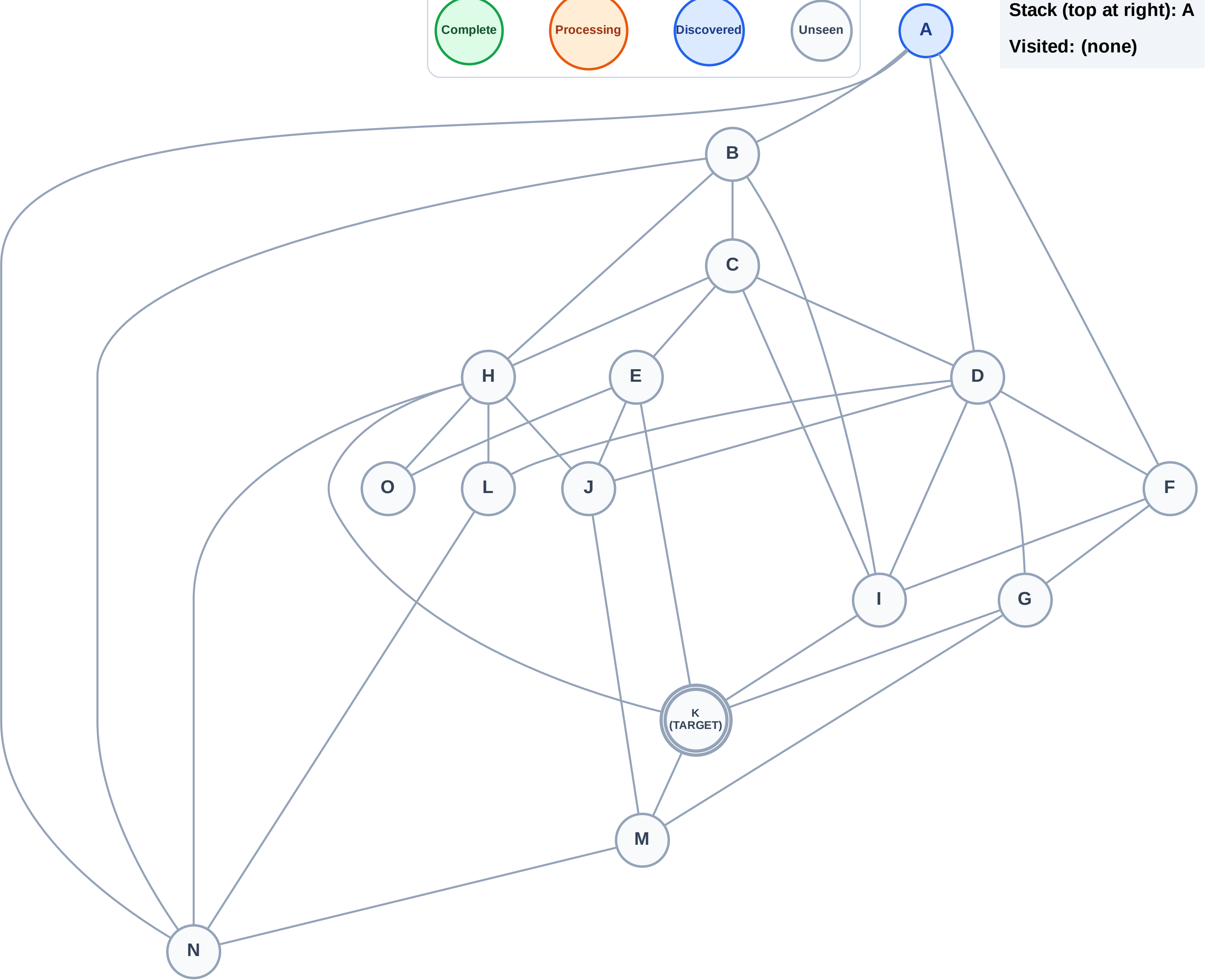
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

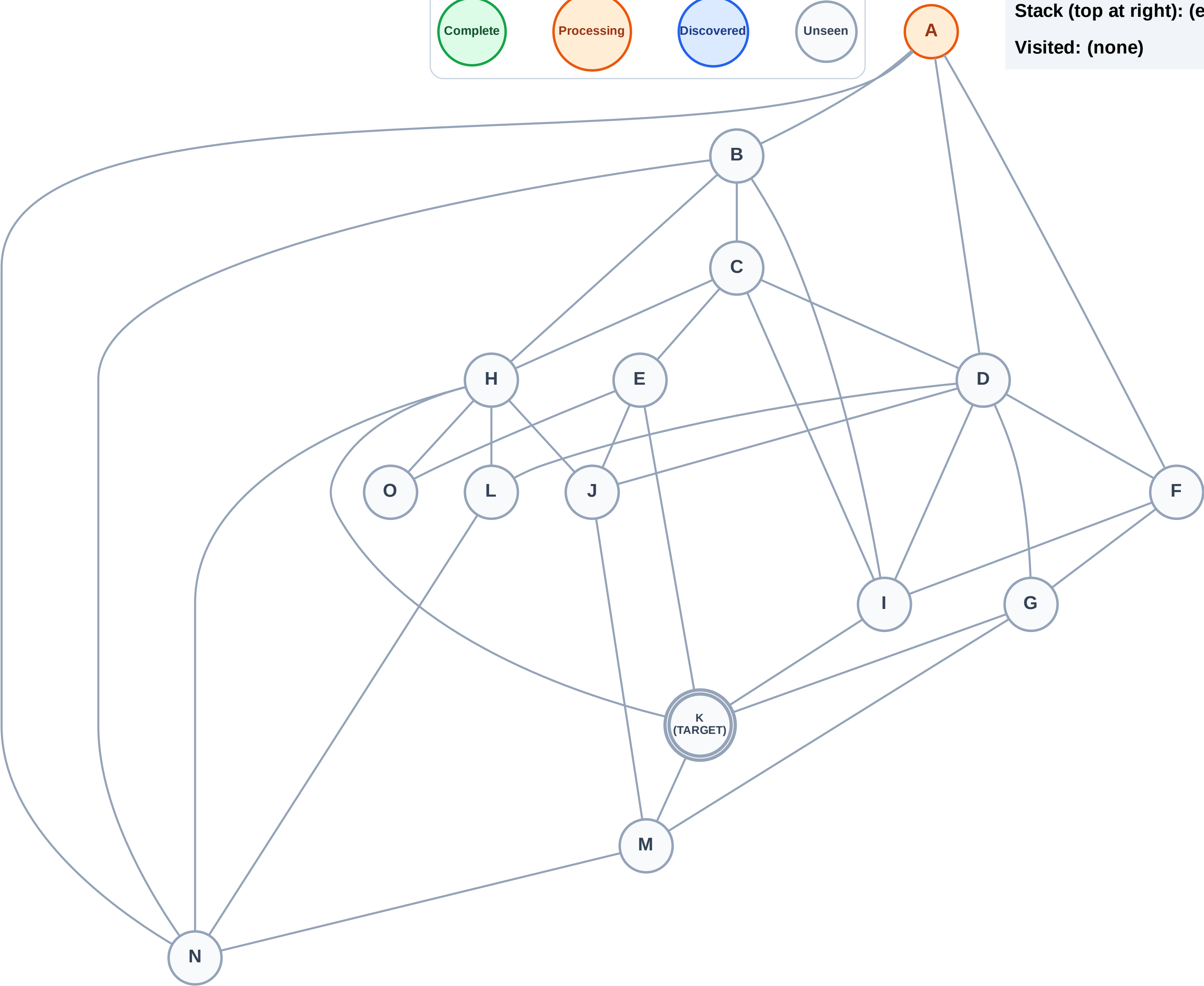
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

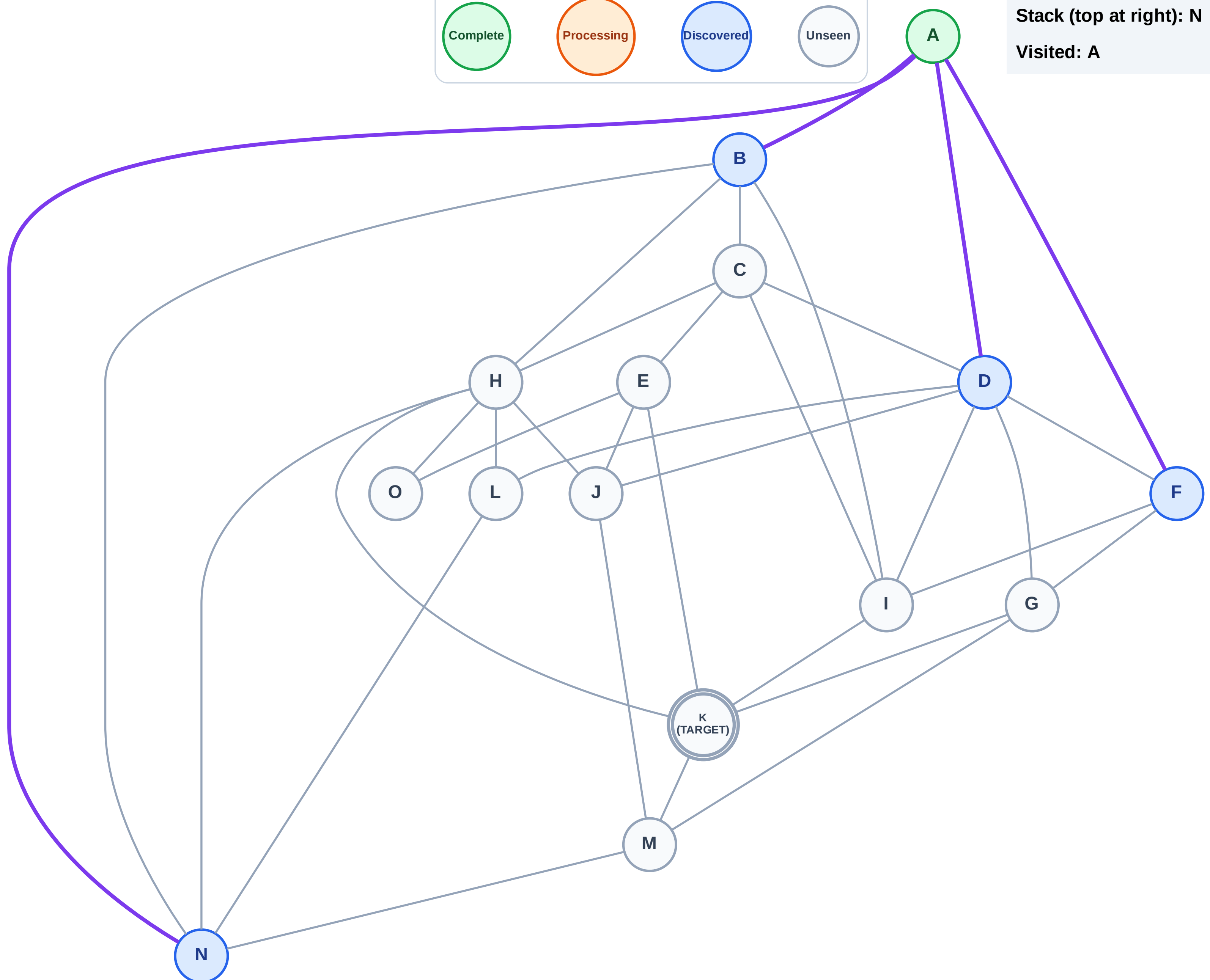
Step 3: Discover N, F, D, B from A

Legend



Stack (top at right): N | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

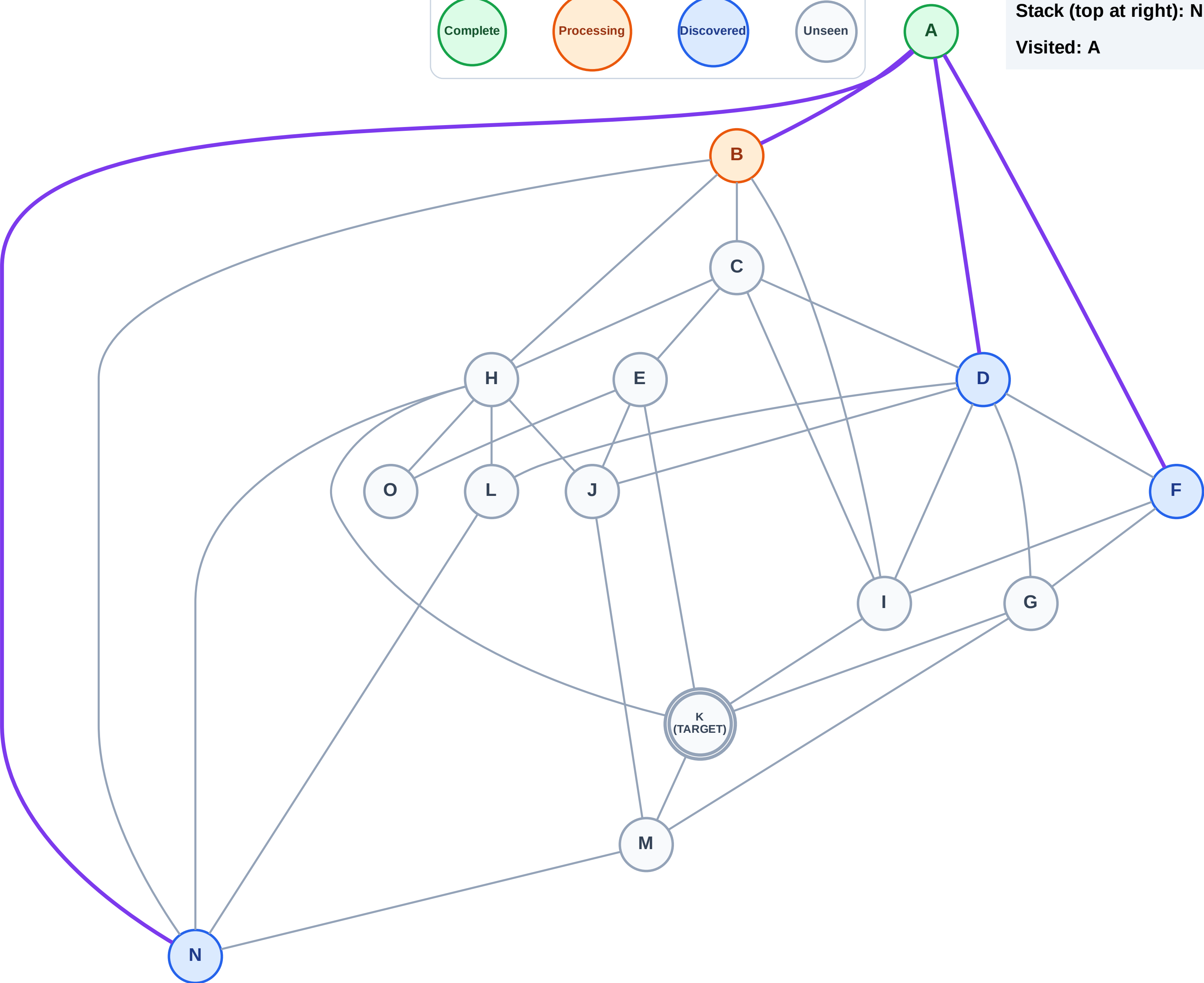
Processing

Discovered

Unseen

Stack (top at right): N | F | D

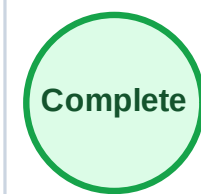
Visited: A



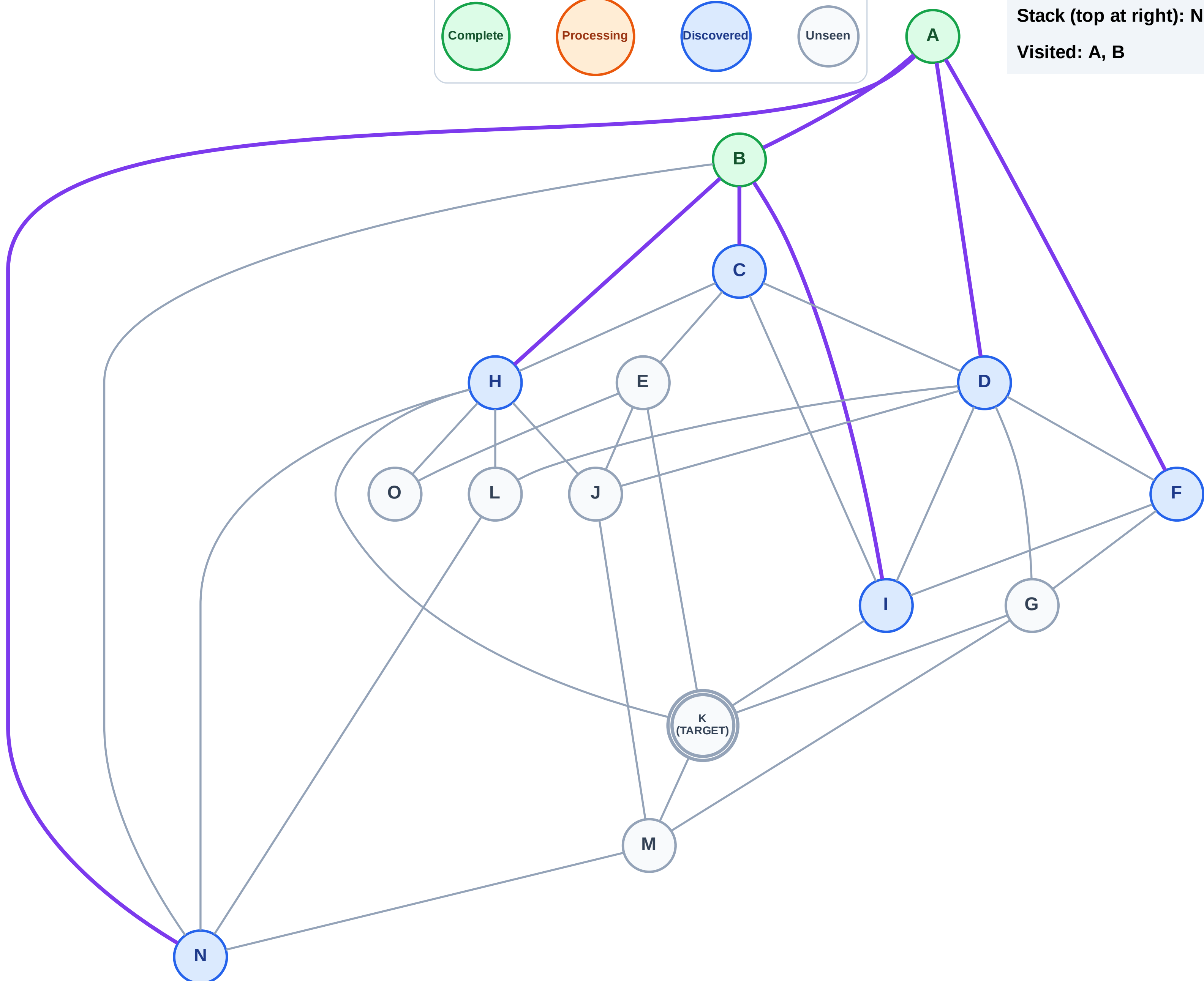
DFS Traversal

Step 5: Discover I, H, C from B

Legend



Stack (top at right): N | F | D | I | H | C
Visited: A, B



DFS Traversal

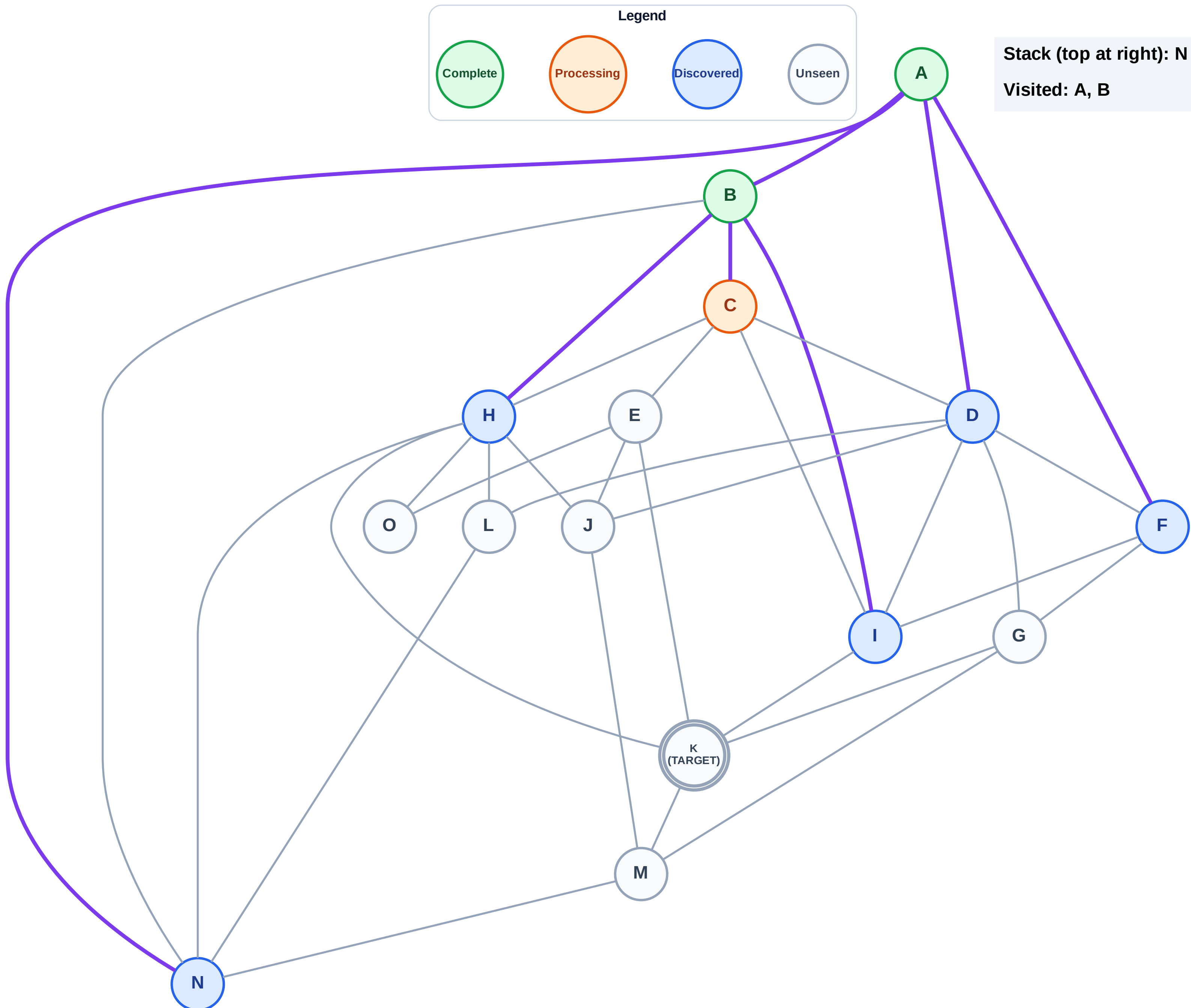
Step 6: Process node C

Legend



Stack (top at right): N | F | D | I | H

Visited: A, B



DFS Traversal

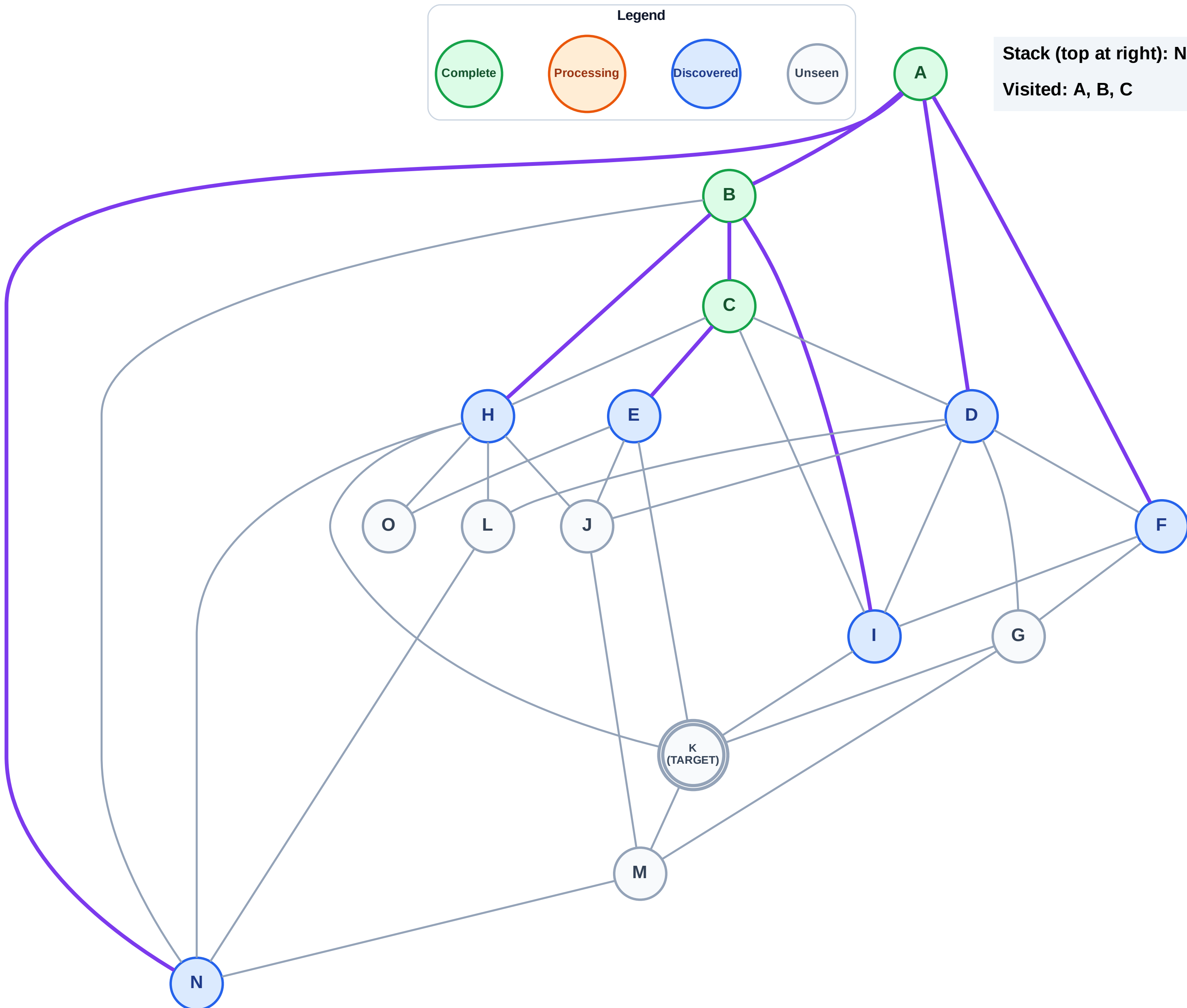
Step 7: Discover E from C

Legend



Stack (top at right): N | F | D | I | H | E

Visited: A, B, C



DFS Traversal

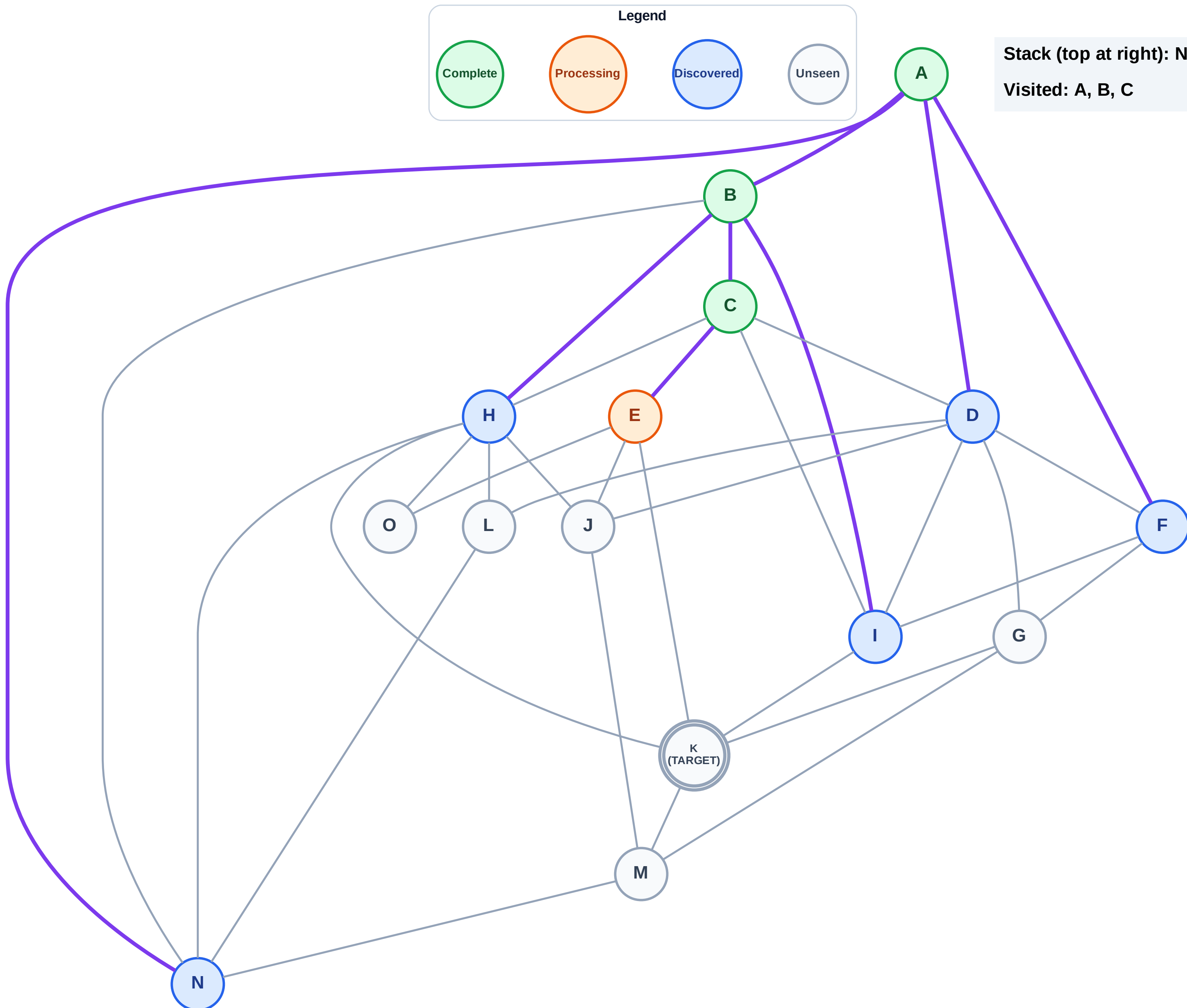
Step 8: Process node E

Legend



Stack (top at right): N | F | D | I | H

Visited: A, B, C



DFS Traversal

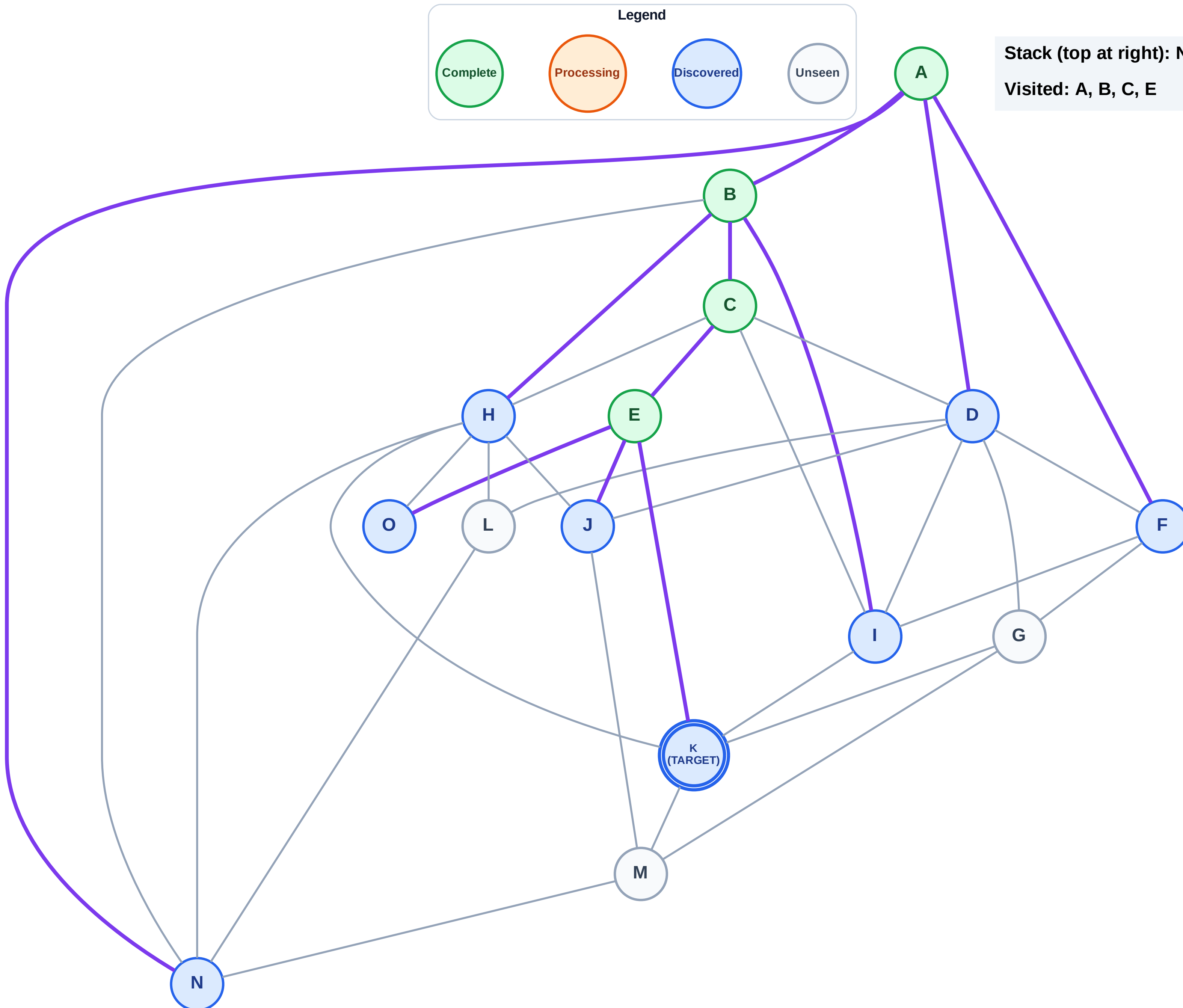
Step 9: Discover O, K, J from E

Legend



Stack (top at right): N | F | D | I | H | O | K | J

Visited: A, B, C, E



Step 10: Process node J

Step 10: Process node J

Legend

Complete

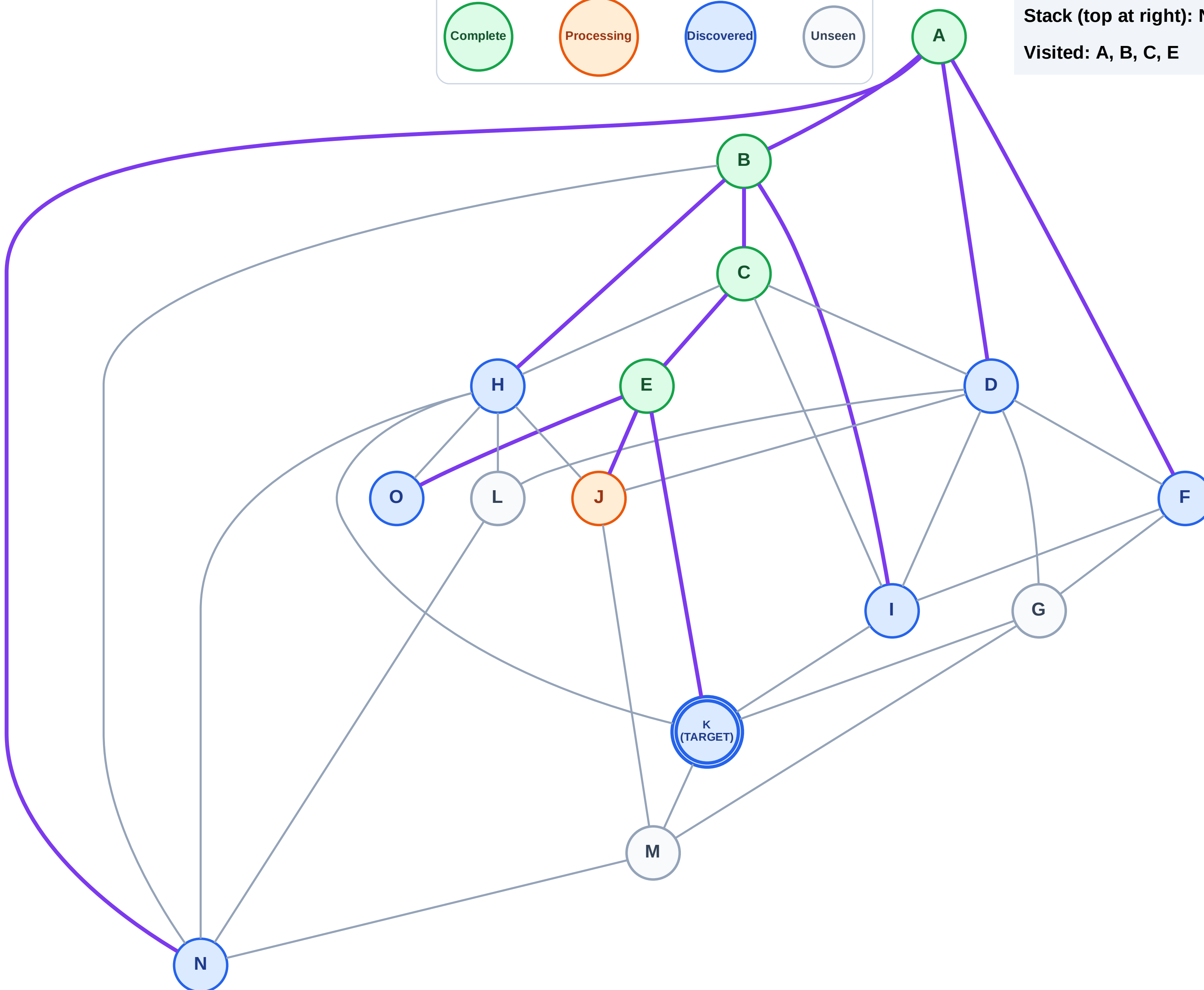
Processing

Discovered

Unseen

Stack (top at right): N | F | D | I | H | O | K

Visited: A, B, C, E



Step 11: Discover M from $\mathcal{D}$

Step 11: Discover M from $\mathcal{D}$

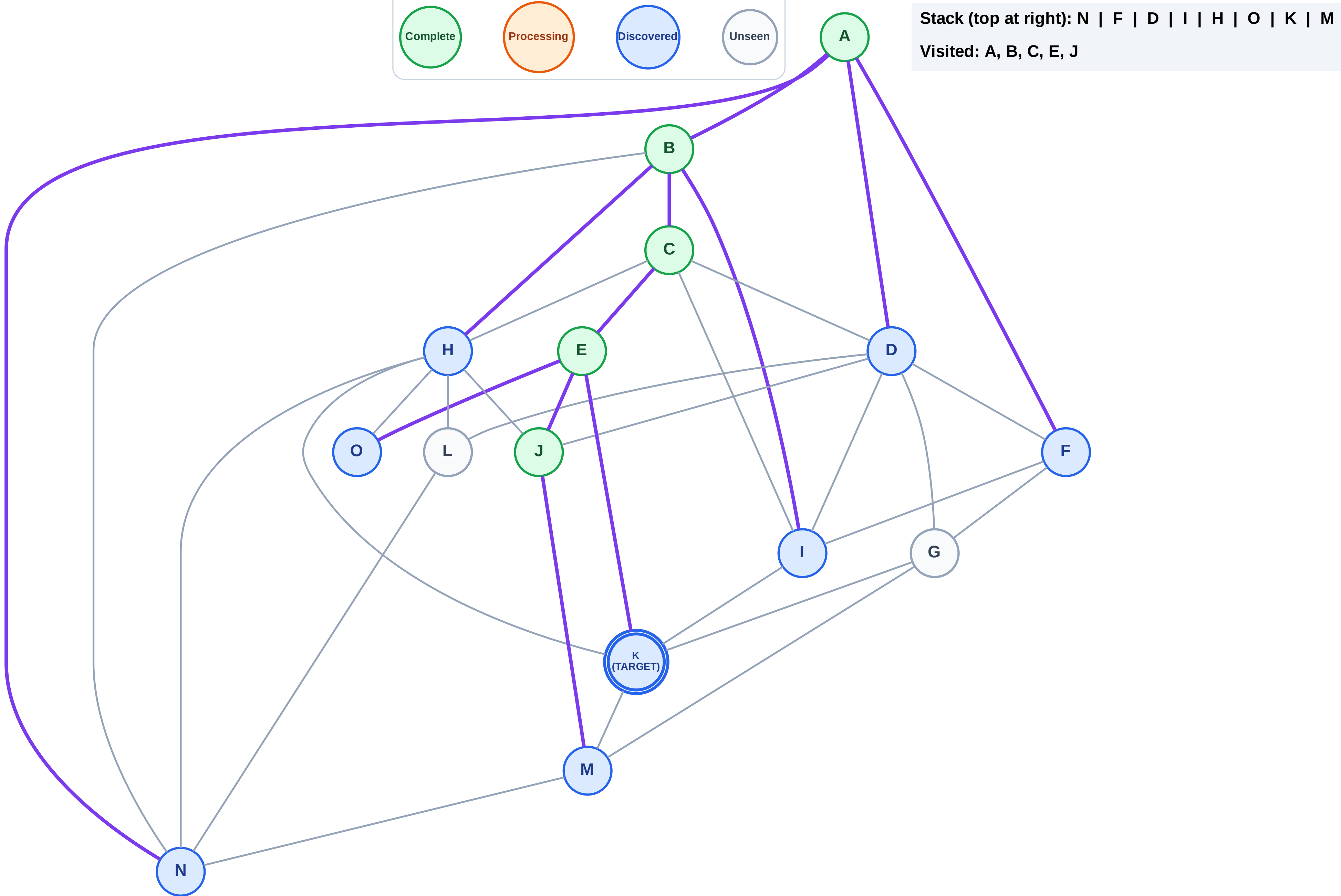
Complete

Processing

Discovered

Unseen

Visited: A, B, C, E, J



Step 12: Process node M

Legend

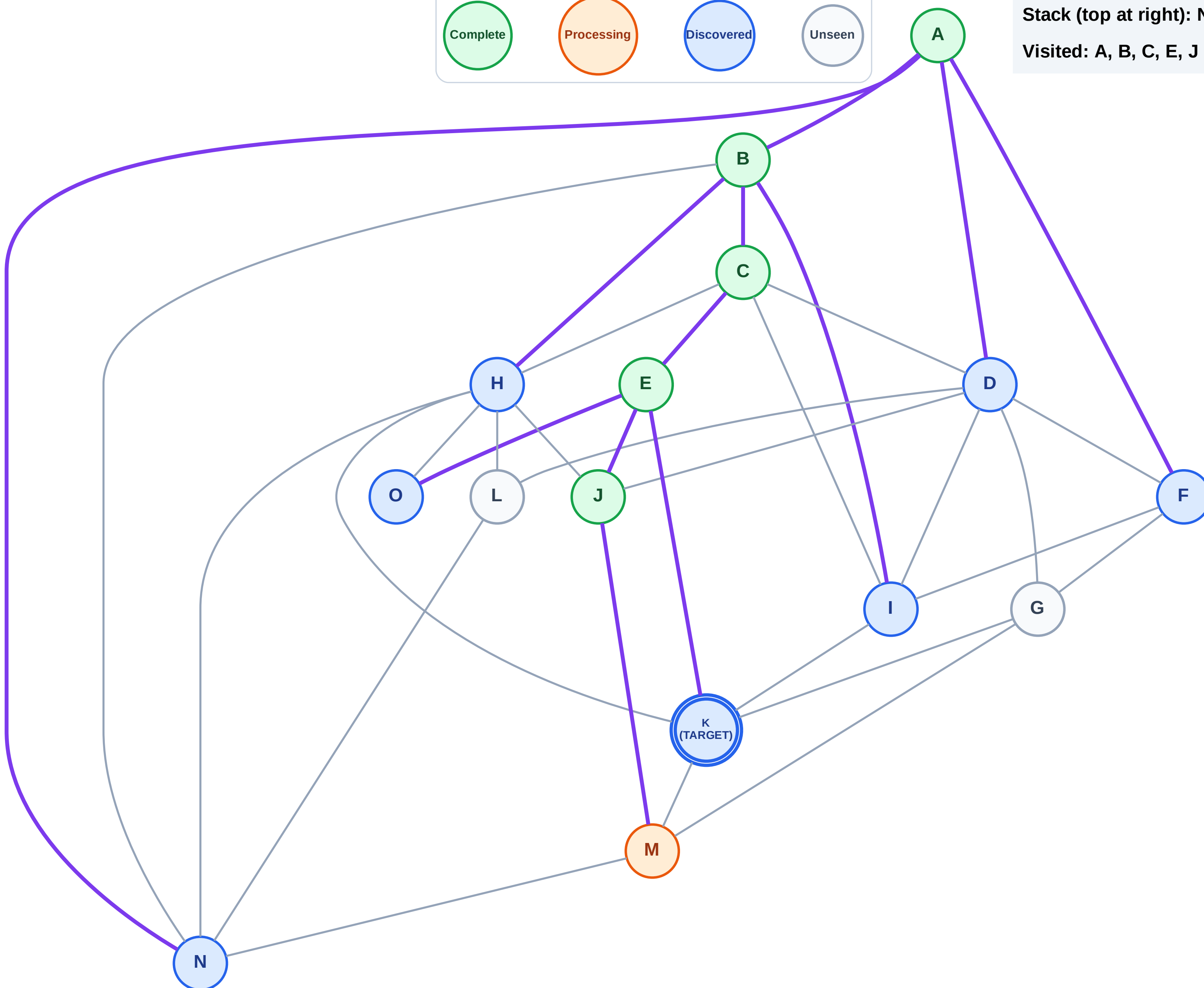
Legend



Complete Processing Discovered Unseen

Unseen

Visited: A, B, C, E, J



DFS Traversal

Step 13: Discover G from M

Legend



Complete



Processing



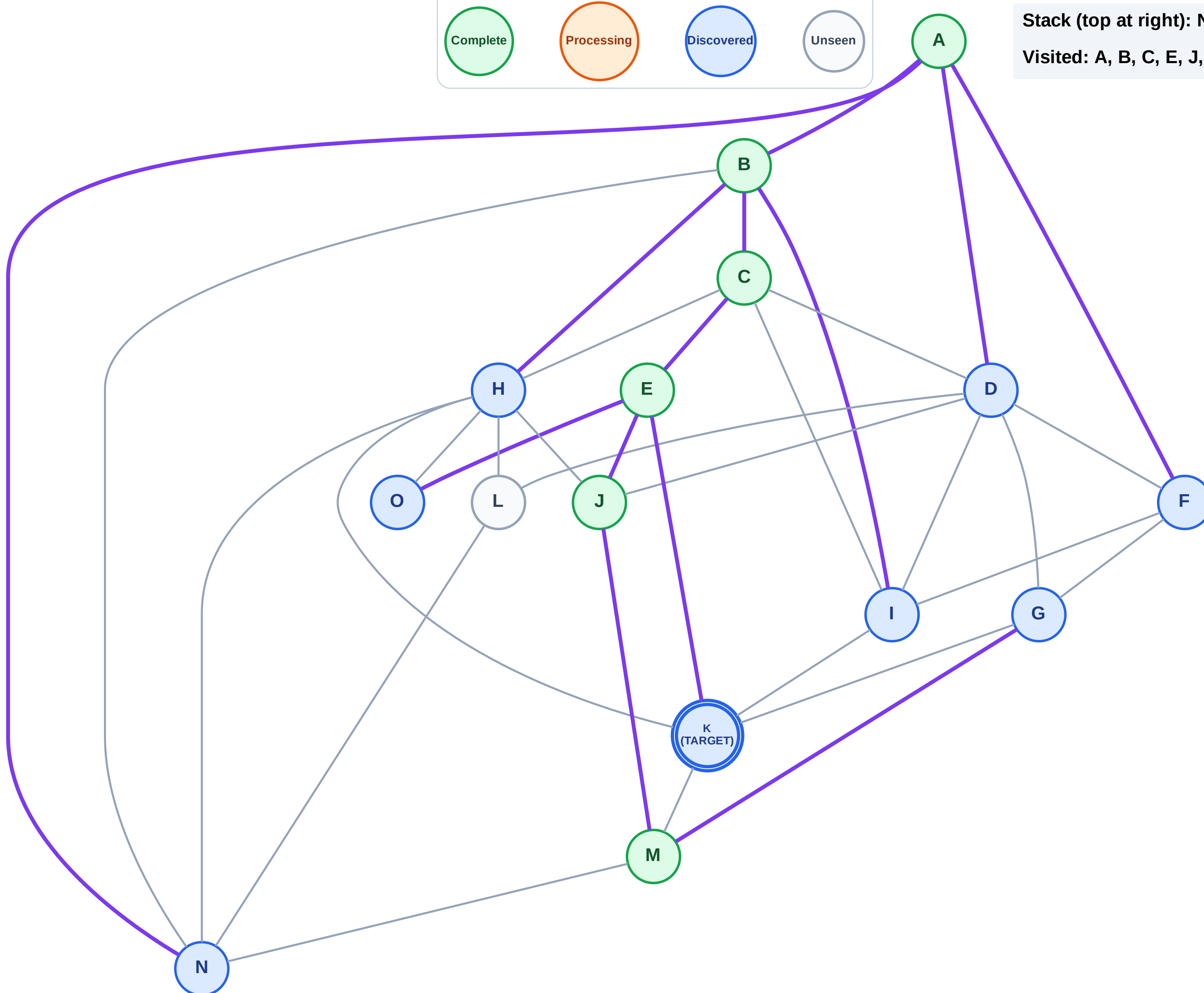
Discovered



Unseen

Stack (top at right): N | F | D | I | H | O | K | G

Visited: A, B, C, E, J, M



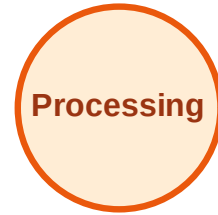
DFS Traversal

Step 14: Process node G

Legend



Complete



Processing



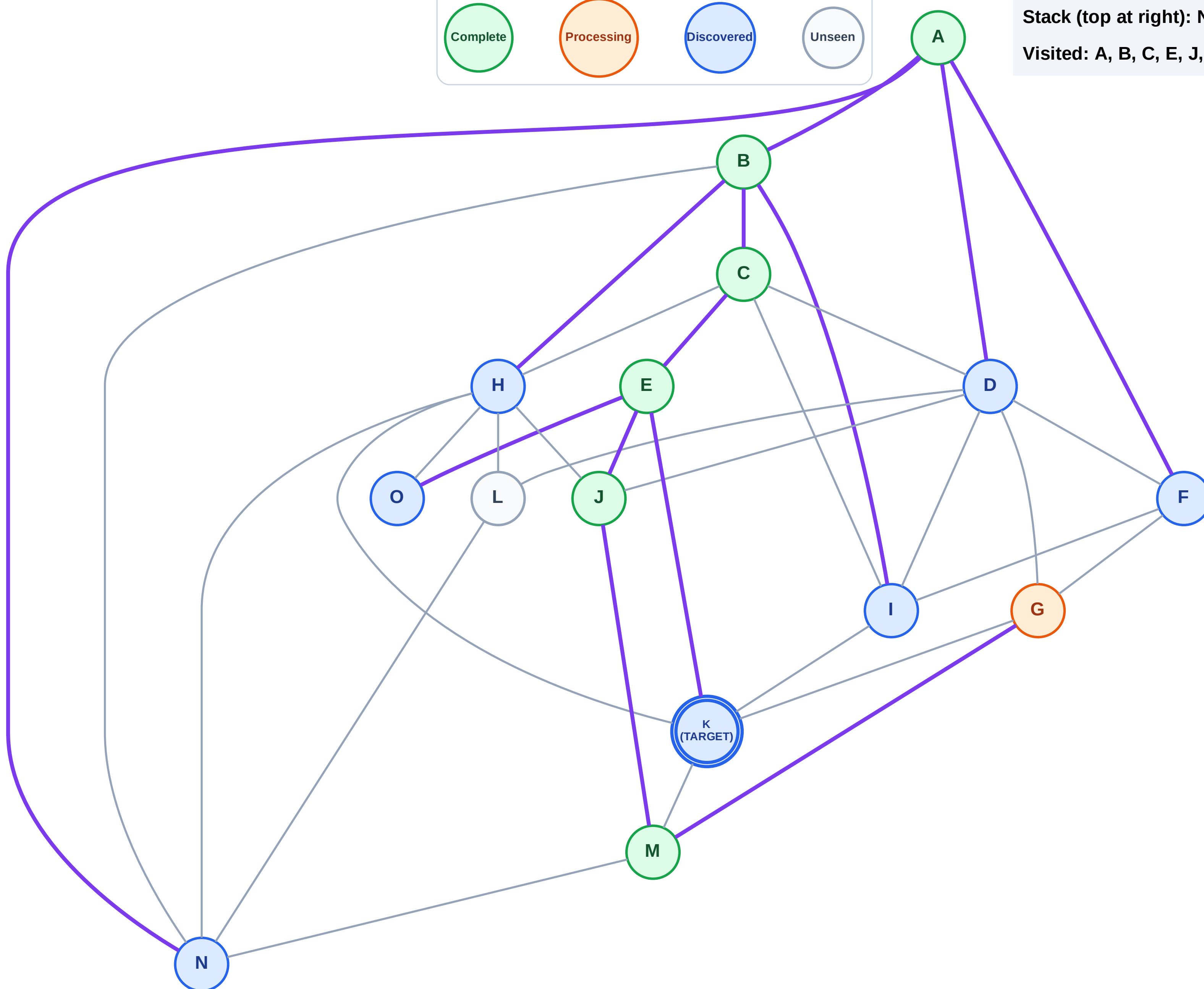
Discovered



Unseen

Stack (top at right): N | F | D | I | H | O | K

Visited: A, B, C, E, J, M



Step 15: No new neighbors from G

Step 15: No new neighbors from G



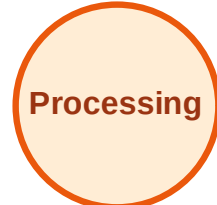
DFS Traversal

Step 16: Process node K

Legend



Complete



Processing



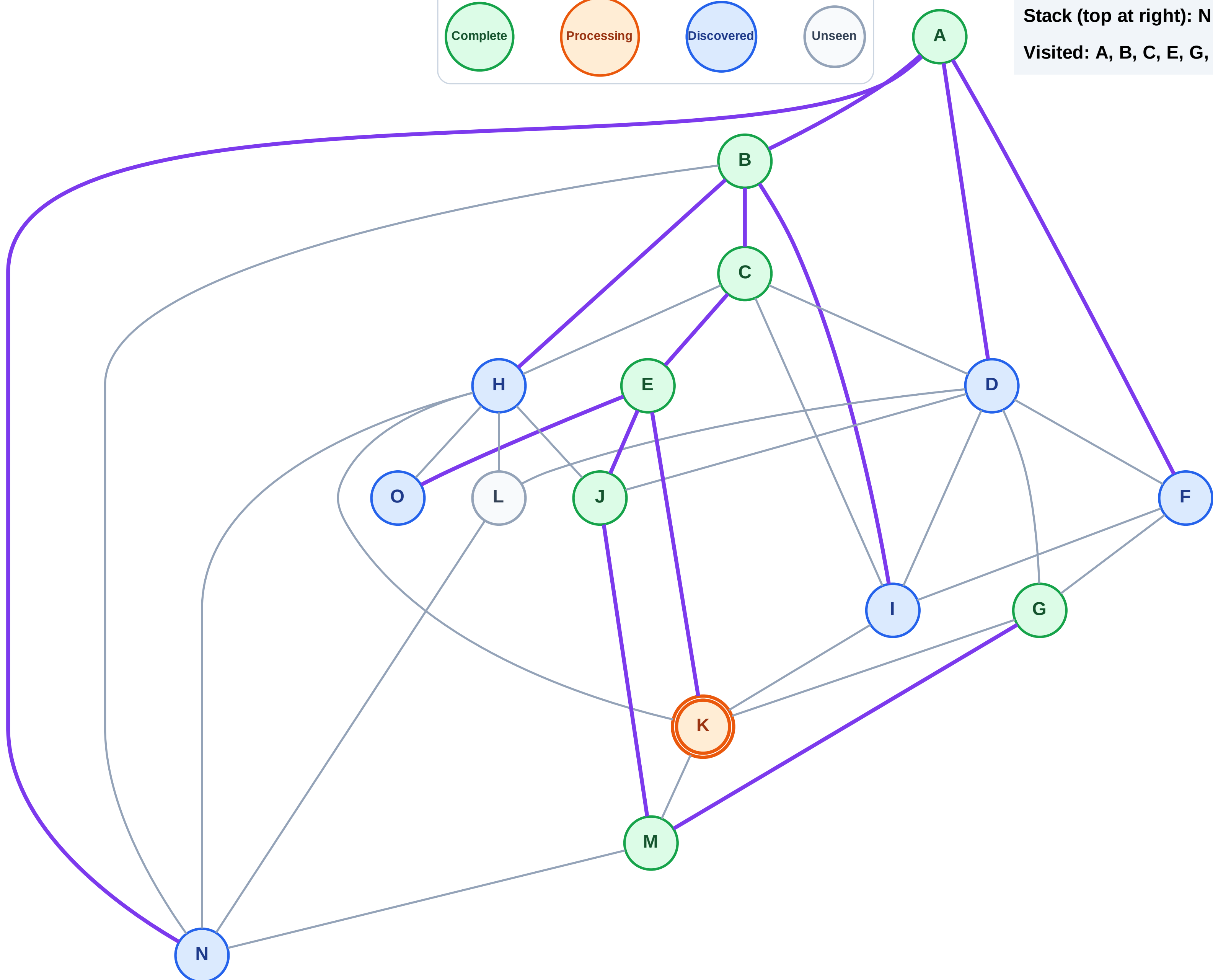
Discovered



Unseen

Stack (top at right): N | F | D | I | H | O

Visited: A, B, C, E, G, J, M



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | F | D | I | H | O

Visited: A, B, C, E, G, J, K, M

